Identification via Heteroskedasticity when Attention is Endogenous

Alistair Macaulay[*]
University of Surrey

July 22, 2026

Abstract

This paper demonstrates that rational inattention can bias structural vector autoregressions identified via heteroskedasticity. Inattentive agents optimally acquire more information during high-volatility regimes, so shock transmission changes with shock variance, which violates the key identifying assumption of constant structural parameters. I characterize the resulting bias and develop two correction methods: one using attention proxies, another exploiting multiple variance regimes. Applying the first of these methods to unconventional monetary policy at the zero lower bound reveals that standard estimates substantially understate the role of expectations in policy transmission, with signaling effects on inflation expectations particularly prominent.

JEL codes: C32, D83, E52

[*]School of Economics, University of Surrey. Email: a.macaulay@surrey.ac.uk. I thank Sarah Duffy, Francesco Fusari, Quinlan Lee, Bartosz Maćkowiak, Joe Marlow, Alexander Meyer-Gohde, Mirela Miescu, Kamila Nowakowicz, Ricardo Nunes, João Santos Silva, Alessio Volpicella, Xueliang Wang, and Yifan Zhang for helpful comments and suggestions.

1 Introduction

Households, firms, and investors acquire more information about economic objects when they have strong incentives to do so. This is the core premise of models of 'rational inattention,' in which agents choose how much information to process about the variables affecting their decision problems, subject to a cost. Following Sims (2003), a large literature has explored how this endogenous information acquisition affects a wide range of macroeconomic dynamics.[1] There is also a growing body of empirical evidence supporting the mechanism.[2]

In this paper I show that endogenous information acquisition also affects the identification of structural vector autoregressions (VARs) via heteroskedasticity. This approach, popularized by Rigobon (2003) and reviewed recently in Lewis (2025), is a commonly used tool for applied macroeconomics. It is often motivated as requiring fewer structural economic assumptions than other identification schemes, as the key assumptions are statistical, not economic. However, once we allow for the possibility of endogenous information acquisition, this ceases to be true. Rational inattention therefore has econometric implications, not just macroeconomic ones.

The key argument is straightforward. Identification via heteroskedasticity exploits changes over time in the variance-covariance matrix of the structural shocks hitting the economy. These changes can be used to identify the transmission of the shocks, *if that transmission is constant across variance regimes*. That is, the mapping from structural innovations to observable variables must remain constant. However, a core lesson of rational inattention models is that when the variance of a shock rises, decision-makers optimally pay more attention to it (Sims, 2003, 2010). At the same time, greater attention affects how shocks transmit to decisions and aggregate outcomes. The 'constant shock transmission' required for identification therefore fails, unless we impose a strong economic assumption that agent information processing does not vary, or does not affect macroeconomic dynamics.

I make three contributions. Theoretically, I show that the presence of rational inattention biases structural VAR estimates identified via heteroskedasticity. Methodologically, I develop two techniques to correct this bias: one using an external proxy for attention, another using observations of three or more variance regimes. Empirically, I apply this bias-corrected identification to assess the role of expectations and attention in the transmission of unconventional monetary policy shocks.

Acknowledging the effects of rational inattention on identification via heteroskedasticity has benefits beyond bias correction. When attention is endogenous, impulse responses to structural shocks differ depending on the variance regime, as that regime determines information choices.

[1] See e.g. Maćkowiak and Wiederholt (2009, 2015); Pasten and Schoenle (2016); Stevens (2020); Song and Stern (2024); Macaulay (2021, 2025), and the review in Maćkowiak et al. (2023).

[2] e.g. Mondria and Quintana-Domeque (2013); Cavallo et al. (2017); Coibion et al. (2018); Roth and Wohlfart (2020); Dean and Neligh (2023); Weber et al. (2025), among many others.

The bias correction methods I propose recover these distinct regime-specific impulse responses. Comparing them therefore reveals how expectations, and information acquisition, affect shock transmission. This insight is not available from other VAR identification schemes that yield a single impulse response per shock.[3]

I apply these insights to the study of unconventional monetary policy, extending the canonical implementation of identification via heteroskedasticity in Wright (2012) to the full zero lower bound (ZLB) period in the US (2008-2015). Wright's identifying assumptions are that (i) the variance of monetary policy shocks is greater on days with scheduled Federal Open Market Committee (FOMC) announcements than on other days; and (ii) the transmission of a unit monetary policy shock is constant over time. Rational inattention calls the second of these into question.

To check if rational inattention is important in this context, I conduct two tests. First, I show that the difference between the covariance matrices of reduced-form innovations in the two regimes has an eigenvalue structure inconsistent with the assumption of constant shock transmission, but which *is* predicted by the model with rational inattention. Second, I use daily Google Trends searches for "Federal Reserve" as a proxy for attention to monetary policy. Search volume increases substantially on FOMC announcement days, regardless of the realized monetary policy shocks that occurred (as identified by Swanson, 2021). This suggests that attention to monetary policy is systematically higher in the high-variance regime, precisely as the theory predicts.

Given these findings, I then use the Google Trends series to implement the first of the correction methods I propose in the earlier part of the paper. This involves using the attention proxy to infer the ratio of structural shock variances across regimes. When combined with an assumption that one variable in the system is unaffected by attention on impact, this pins down the rotation among observationally equivalent impact vectors. Intuitively, knowing how much more agents pay attention in the high-variance regime reveals how much of the observed covariance difference is due to changing variances versus changing transmission, allowing us to separate the two.

The second correction method does not require a proxy for attention, and so is more conservative in that dimension. However, it does require three or more observable variance regimes. The insight is that with a single shock of interest, the change in impulse responses between regimes 1 and 2 is collinear with the change between regimes 2 and 3, because both shifts occur due to a one-dimensional change in attention to the shock. This collinearity provides an extra restriction that can be used to disentangle the structural parameters of the model. Since the monetary policy application has only two clearly defined variance regimes, I use the attention-proxy method there, and provide an alternative application of this second method in Appendix F.

The results imply that the updating of beliefs about the stance of monetary policy, through

[3]This includes recursive identification (e.g. Christiano et al., 1999), long-run restrictions (e.g. Blanchard and Quah, 1989), as well as approaches using external instruments (e.g. Gürkaynak et al., 2005b).

attentive agents, is of central importance to the transmission of unconventional monetary policy shocks. Uncorrected impulse responses from conventional identification via heteroskedasticity are strongly distorted by endogenous information acquisition, often lying outside the range of the corrected regime-specific responses. These uncorrected estimates imply greater persistence in the effects of unconventional monetary policy on Treasury and corporate bond yields than is observed after the bias correction, and they miss a large short-term swing in breakeven inflation rates.

Specifically, I find that the immediate impact of policy on Treasury yields is similar across regimes, consistent with mechanical portfolio rebalancing that operates independently of beliefs. However, the *persistence* of these effects differs markedly by regime: after 50 business days, the effect on the 10-year yield is approximately half as large in the low-attention regime as in the high-attention regime. Without the updates to expectations facilitated by high attention, the effects of unconventional policy on the yield curve are largely transitory. High-grade corporate bond yields follow similar patterns, indicating that they are indeed close substitutes for long-term Treasuries, as argued by Krishnamurthy and Vissing-Jorgensen (2011) and others.

For riskier corporate bonds, however, attention acts as a powerful short-run amplifier. BAA yields respond 25% more strongly on impact in the high-attention regime, consistent with a credit channel where information about monetary policy causes investors to update default expectations.

Finally, the bias correction reveals pronounced effects of unconventional policy on inflation expectations. Models of the signaling channel of unconventional monetary policy, such as Eggertsson and Woodford (2003), predict that inflation expectations should increase strongly after expansionary policy shocks. However, uncorrected impulse responses, such as those in Wright (2012), display only muted effects on market-based expectations. I find that this is driven by a strong 'rotation' in the low-attention regime, in which medium-term breakeven inflation rises but longer-term breakevens fall, plausibly reflecting mechanical or liquidity-driven distortions. When attention is high, expectation effects offset this, delivering small positive effects overall. The effect of unconventional policy on inflation expectations is therefore substantially stronger than raw breakevens data suggest: the small positive effects on forward breakevens in the high-attention regime include a large negative mechanical effect, which is offset by expectations updating.

These findings therefore add new evidence to the view that signaling effects are a dominant driver of unconventional monetary policy transmission.

Related literature. This paper principally contributes to the literatures on rational inattention and the identification of structural VARs using heteroskedasticity. In the former, a large literature since Sims (2003) has shown that endogenous information acquisition has important effects across macroeconomics and finance, including in price setting (Maćkowiak and Wiederholt, 2009; Pasten and Schoenle, 2016), consumption (Tutino, 2013), business cycles (Maćkowiak and Wiederholt, 2015), monetary policy (Paciello and Wiederholt, 2014), misallocation (Gondhi, 2023), financial markets (Kacperczyk et al., 2016), labor markets (Acharya and Wee, 2020; Ellison and Macaulay, 2021), and household finance (Lei, 2019; Macaulay, 2021, 2025). Empirical evidence has been found in experiments (Dean and Neligh, 2023), surveys (Roth and Wohlfart, 2020; Link et al., 2023), and VARs (Geiger and Scharler, 2021). I extend this literature by showing that endogenous attention also affects macroeconometric analysis, specifically structural VARs identified via heteroskedasticity.

This method of identifying structural VARs was popularized by Rigobon (2003) and Rigobon and Sack (2003, 2004). While these consider discrete observable variance regimes, the method has since been extended to other forms of heteroskedasticity, including Markov-switching models (Lanne et al., 2010), smooth volatility transitions (Lütkepohl and Netšunajev, 2017), and other parametric volatility processes (e.g. Normandin and Phaneuf, 2004).[4] Applications include estimating the spillover effects of sovereign bonds (Rigobon, 2003), cross-border financial market linkages (Ehrmann et al., 2011), monetary policy (Rigobon and Sack, 2003; Wright, 2012; Nakamura and Steinsson, 2018; Schlaak et al., 2023; Acosta, 2026), fiscal multipliers (Lewis, 2021), macro-financial feedback effects (Brunnermeier et al., 2021), and pandemic shocks (Miescu and Rossi, 2021). I show that rational inattention, now a relatively common assumption in macroeconomic theory, can bias structural VARs identified in this way if my correction methods are not applied. Although for clarity I focus on discrete variance regimes and a single shock of interest, the core mechanism extends to more general cases: attention responds to volatility, and this causes the key constant-transmission identifying assumption to fail.

The rational inattention mechanism I study causes the matrix governing the impact effects of structural shocks to vary with the variance regime. Bacchiocchi and Fanelli (2015), Bacchiocchi et al. (2018), and Angelini et al. (2019) also consider variation in this impact matrix, though they allow it to vary in a general way, and thus have to provide extra restrictions to identify the model parameters. As Brenna et al. (2023) point out, these extra restrictions are often difficult to defend. My approach differs in two important ways. First, rather than a general relaxation of the constant-impact matrix assumption, rational inattention provides a concrete economic reason for time-varying transmission, enabling both testing and bias correction. Second, the correction recovers distinct impulse responses under varying levels of attention, which reveals how attention affects the transmission of macroeconomic shocks. This insight is unavailable from approaches that treat impact-matrix variation as a nuisance, or other structural VAR identification schemes.

My application follows from Wright (2012) and Nakamura and Steinsson (2018), who similarly identify monetary policy shocks by assuming higher variance on FOMC announcement days.

[4] See Lütkepohl and Netšunajev (2015) and Lewis (2025) for surveys of possible models. More broadly, the approach also relates to other forms of 'statistical identification,' including that from non-Gaussianity (see Jarociński, 2024, and the references therein), which is discussed in greater detail in Appendix A.2.

More broadly, this exercise contributes to the wider literature on unconventional monetary policy (see reviews in Dell'Ariccia et al. (2018) and Kuttner (2018)). Within this, many papers argue theoretically that expectations play an important role in the transmission of unconventional monetary policy (e.g. Iovino and Sergeyev, 2023; Bhattarai et al., 2023), and find empirical evidence consistent with this (e.g. Bauer and Rudebusch, 2014; Boneva et al., 2016). I provide direct evidence of expectational effects: the bias correction recovers impulse responses to unconventional monetary policy shocks under high and low attention, and thus under different degrees of expectation updating.

2 Identification via Heteroskedasticity: Baseline Framework

Consider a structural VAR(p):

$$\mathbf{A}(L)\mathbf{y}_t = \mathbf{B}\boldsymbol{\varepsilon}_t, \tag{1}$$

where $\mathbf{y}_t$ is an $n \times 1$ vector of observable variables, $\mathbf{A}(L) = \mathbf{I}_n - \mathbf{A}_1 L - \mathbf{A}_2 L^2 - \cdots - \mathbf{A}_p L^p$ is a matrix polynomial in the lag operator, $\mathbf{B}$ is an $n \times n$ invertible matrix mapping structural shocks to observables, and $\boldsymbol{\varepsilon}_t$ is an $n \times 1$ vector of structural shocks. Structural shocks are such that $\mathbb{E}[\boldsymbol{\varepsilon}_t] = 0$ and $\mathbb{E}[\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t'] = \boldsymbol{\Sigma}_\varepsilon$, where $\boldsymbol{\Sigma}_\varepsilon$ is diagonal, implying the structural shocks are mutually uncorrelated.

The reduced-form representation

$$\mathbf{A}(L)\mathbf{y}_t = \mathbf{u}_t, \tag{2}$$

uses $\mathbf{u}_t = \mathbf{B}\boldsymbol{\varepsilon}_t$ to denote reduced-form innovations, which have the covariance matrix

$$\boldsymbol{\Omega} = \mathbb{E}[\mathbf{u}_t \mathbf{u}_t'] = \mathbf{B}\boldsymbol{\Sigma}_\varepsilon \mathbf{B}'. \tag{3}$$

The fundamental identification problem is that there are infinitely many combinations of the structural matrices $\mathbf{B}, \boldsymbol{\Sigma}_\varepsilon$ that could rationalize the observed reduced-form covariance matrix $\boldsymbol{\Omega}$. Without further restrictions, the structural parameters are not separately identified.

Identification via heteroskedasticity. The heteroskedasticity-based approach, following Rigobon (2003), exploits variation in shock variances across regimes. Suppose there exist $K \geq 2$ variance regimes indexed by $s \in \{1, \ldots, K\}$. In regime s, structural shocks have a covariance matrix given by

$$\boldsymbol{\Sigma}_\varepsilon^{(s)} = \text{diag}(\sigma_{1,s}^2, \sigma_{2,s}^2, \ldots, \sigma_{n,s}^2), \tag{4}$$

where the variances $\sigma_{j,s}^2$ may differ across regimes. Throughout, I will use a superscript (s) to denote matrices specific to regime s. The key identifying assumption is:

Assumption 1 (Structural Invariance) *The impact matrix* $\mathbf{B}$ *is constant across variance regimes:*

$$\mathbf{B}^{(s)} = \mathbf{B} \quad \forall s \in \{1, \ldots, K\}. \tag{5}$$

Example: monetary policy on FOMC days. A canonical application is Wright (2012), who identifies monetary policy shocks by comparing FOMC announcement days (high variance for monetary policy shocks) to non-announcement days (low variance). In this context, Assumption 1 implies that the responses to monetary policy shocks are the same on FOMC and non-FOMC days; only the variance of the monetary policy shock is different.

Identification procedure. Under Assumption 1, the reduced-form covariance in regime s is

$$\mathbf{\Omega}^{(s)} = \mathbf{B}\mathbf{\Sigma}_{\varepsilon}^{(s)}\mathbf{B}'. \tag{6}$$

Identification exploits the differences across regimes. Taking differences between regime s and regime 1, we obtain

$$\mathbf{\Omega}^{(s)} - \mathbf{\Omega}^{(1)} = \mathbf{B}\left(\mathbf{\Sigma}_{\varepsilon}^{(s)} - \mathbf{\Sigma}_{\varepsilon}^{(1)}\right)\mathbf{B}'. \tag{7}$$

If the variance changes are concentrated in a single shock (say shock 1), then $\mathbf{\Sigma}_{\varepsilon}^{(s)} - \mathbf{\Sigma}_{\varepsilon}^{(1)}$ has rank one, and we can write

$$\mathbf{\Omega}^{(s)} - \mathbf{\Omega}^{(1)} = (\sigma_{1,s}^2 - \sigma_{1,1}^2)\mathbf{b}_1\mathbf{b}_1', \tag{8}$$

where $\mathbf{b}_1$ is the first column of $\mathbf{B}$. The spectral decomposition of $\mathbf{\Omega}^{(s)} - \mathbf{\Omega}^{(1)}$ can then be used to recover $\mathbf{b}_1$ up to sign and scale. The sign ambiguity is typically resolved by convention (e.g., a contractionary monetary policy shock raises the policy rate). The scale is pinned down by a normalization, such as fixing the variance of the shock in one regime or fixing one element of $\mathbf{b}_1$. Wright (2012), for example, uses the normalization that $(\sigma_{1,s}^2 - \sigma_{1,1}^2) = 1$.

The key point of this paper is that when agents engage in optimal information acquisition, Assumption 1 will fail. If the variance of structural shocks changes from one regime to another, this will cause agents to change their information choices, which in turn will affect how they react to shocks, leading to a change in $\mathbf{B}$. In other words, the response to structural shocks differs systematically across these regimes precisely *because* the variance differs.

3 Rational Inattention and Regime-Dependent Transmission

This section introduces rational inattention into the SVAR framework and shows how this biases estimates identified through heteroskedasticity. For simplicity, I focus here on the standard case in which the structural shock of interest is i.i.d. Normal.

3.1 Environment

Denote the structural shock of interest as $\varepsilon_{1,t}$. In variance regime s, this shock is distributed according to $\varepsilon_{1,t} \sim \mathcal{N}(0, \sigma_s^2)$. Without loss of generality I assume regime $s = K$ has the greatest variance, i.e. $\sigma_K^2 > \sigma_{s \neq K}^2$. In the monetary policy example from Wright (2012) discussed above, $\varepsilon_{1,t}$ is the monetary policy shock, the number of regimes is $K = 2$, and s is equal to 2 (high variance) on FOMC days and 1 (low variance) on non-FOMC days.

I now make two assumptions which are irrelevant in existing treatments of identification via heteroskedasticity, but which are important when introducing information frictions into the model.

Assumption 2 (Transmission through Expectations) *The effect of shock $\varepsilon_{1,t}$ on the vector of outcomes $\mathbf{y}_t$ is:*

$$\mathbf{A}(L)\mathbf{y}_t = \boldsymbol{\alpha}_1 \varepsilon_{1,t} + \boldsymbol{\beta}_1 \bar{\mathbb{E}}_t[\varepsilon_{1,t}] + \textit{other shocks}. \tag{9}$$

where $\bar{\mathbb{E}}_t[\varepsilon_{1,t}]$ is the average shock perception across a continuum of agents, defined in Section 3.4.

The vector $\boldsymbol{\alpha}_1$ captures transmission channels such as direct asset price effects or mechanical balance-sheet adjustments that operate independently of agent beliefs. In turn, $\boldsymbol{\beta}_1$ captures transmission channels that depend on agents' perceptions of the shock. In the monetary policy application, $\boldsymbol{\beta}_1$ may come from consumption, investment, or pricing decisions which require households and firms to form beliefs about the policy stance. While equation (9) is intentionally left general, I provide a microfoundation in an asset-pricing context in Appendix A.1.

I show below that the average shock perception $\bar{\mathbb{E}}_t[\varepsilon_{1,t}]$ is a deterministic function of the true shock $\varepsilon_{1,t}$, so Assumption 2 does not imply a departure from the structural VAR form in (1).

Assumption 3 (Known Variance Regimes) *Agents observe the variance regime s before forming expectations and making decisions.*

Assumption 3 is natural for applications like FOMC announcements, where the schedule is publicly known. This assumption is crucial: if agents did not know the variance regime ex ante, attention would not condition on volatility, and my critique would not apply.

3.2 Information Structure

Agents do not observe $\varepsilon_{1,t}$ directly. Instead, I follow the rational inattention literature and assume that agents can pay to observe possibly noisy signals about $\varepsilon_{1,t}$, where more precise signals come with a greater cost. Agents observe all other structural shocks directly and without error.[5]

Specifically, each agent i chooses a signal structure, which consists of a signal space $\mathcal{Z}$ and a conditional distribution $\pi(z|\varepsilon_{1,t})$ over signals $z \in \mathcal{Z}$ given the realization of the shock. The agent then takes an action $a_{i,t}$ after observing the signal realization. As is common in the rational inattention literature,[6] I assume a quadratic objective function, such that agent i wishes to minimize the expected squared deviation of their action from the realized shock:

$$\min_{a_{i,t}} \mathbb{E}\left[(a_{i,t} - \varepsilon_{1,t})^2 \,\middle|\, z_{i,t}, s\right]. \tag{10}$$

This expectation is formed conditional on the agent's information set, which consists of their idiosyncratic signal $z_{i,t}$ and, via Assumption 3, the regime s. Given this information, the optimal action is the conditional expectation $a^*_{i,t} = \mathbb{E}[\varepsilon_{1,t}|z_{i,t}, s]$, and the minimized loss is the posterior variance $\text{Var}(\varepsilon_{1,t}|z_{i,t}, s)$.

Having found the agent's payoff at the optimal action, conditional on a given signal, I now turn to the choice of signal structure. Following Sims (2003), I impose that the cost of a signal structure is proportional to the mutual information between the shock and the signal, defined as

$$\mathcal{I}(\varepsilon_{1,t}; z) = H(\varepsilon_{1,t}) - \mathbb{E}_z\left[H(\varepsilon_{1,t}|z)\right], \tag{11}$$

where $H(\cdot)$ denotes Shannon entropy. Intuitively, mutual information measures the expected reduction in uncertainty about $\varepsilon_{1,t}$ from observing the signal. More precise signals that imply larger expected shrinkage from prior beliefs to posteriors are therefore more costly for the agent to process.

The agent's information-choice problem is therefore to choose a signal structure to minimize the sum of the expected loss from equation (10) and information costs. Letting $\theta > 0$ denote the marginal cost of information, this problem can be written:

$$\min_{\pi(z|\varepsilon_{1,t})} \mathbb{E}\left[\text{Var}(\varepsilon_{1,t}|z, s)\right] + \theta \cdot \mathcal{I}(\varepsilon_{1,t}; z), \tag{12}$$

3.3 Optimal Attention Choice

Given a quadratic objective function and Gaussian uncertainty, it can be shown (Sims, 2003) that the optimal signal structure consists of a single noisy signal of the form

$$z_{i,t} = \varepsilon_{1,t} + \eta_{i,t}, \quad \eta_{i,t} \sim \mathcal{N}(0, 1/\kappa_{i,t}), \tag{13}$$

[5] This is assumed to clearly isolate the role of attention to the shock of interest. If information costs are additively separable across variables, as in e.g. Afrouzi and Yang (2021), then relaxing this assumption is inconsequential.

[6] See e.g. Maćkowiak and Wiederholt (2009, 2015), who show that this can be motivated as a quadratic approximation to richer objective functions.

where $\kappa_{i,t} > 0$ is the signal precision. With this in hand, the posterior expectation of $\varepsilon_{1,t}$ can be written as

$$\mathbb{E}(\varepsilon_{1,t}|z_{i,t}, s) = \tau_{i,t} z_{i,t}, \quad \text{where} \quad \tau_{i,t} = \frac{\sigma_s^2}{\sigma_s^2 + 1/\kappa_{i,t}}. \tag{14}$$

Here $\tau_{i,t}$ is the posterior weight on the signal, which is monotonically increasing in the signal-to-noise ratio $\kappa_{i,t}\sigma_s^2$. Since $\tau_{i,t}$ is also monotonically increasing in the amount of information processed $\mathcal{I}(\varepsilon_{1,t}; z)$, I will refer to it as a measure of the agent's attention.

The agent problem (12) can then be reduced to the choice of precision $\kappa_{i,t}$, or equivalently attention $\tau_{i,t}$. Evaluating the posterior variance Var$(\varepsilon_{1,t}|z_{i,t}, s)$ and the mutual information $\mathcal{I}(\varepsilon_{1,t}; z)$ with the signal structure in (13), we can write the agent's information-choice problem as

$$\min_{\tau_{i,t}\in[0,1]} \left\{ \sigma_s^2(1-\tau_{i,t}) + \theta \cdot \frac{1}{2} \log\left(\frac{1}{1-\tau_{i,t}}\right) \right\}. \tag{15}$$

Lemma 1 (Optimal Attention) *The optimal attention for each agent in period t is:*

$$\tau_s^* = \begin{cases} 1 - \frac{\theta}{2\sigma_s^2} & \textit{if } \sigma_s^2 > \frac{\theta}{2} \\ 0 & \textit{otherwise} \end{cases}, \tag{16}$$

which is strictly increasing in σ_s^2 for $\sigma_s^2 > \theta/2$.

Proof. Appendix B.1. ■

The key implication of Lemma 1 is that optimal attention is increasing in the variance of the underlying shock:

$$\sigma_j^2 > \sigma_k^2 \implies \tau_j^* \geq \tau_k^* \tag{17}$$

for any pair of regimes j, k. The inequality in attention is strict whenever $\sigma_j^2 > \frac{\theta}{2}$.

This is a very standard feature of models of rational inattention. Agents allocate more attention to monitoring monetary policy when monetary policy shocks are more volatile, because large volatility means that agents would make larger expectational errors if they did not process information. The key argument of this paper is that this standard mechanism has consequences for the identification of structural VARs via heteroskedasticity, because it links variance regimes with shock transmission.

3.4 Aggregation with Idiosyncratic Signals

With a continuum of agents $i \in [0, 1]$ receiving idiosyncratic signals, the aggregate expectation is

$$\bar{\mathbb{E}}_t[\varepsilon_{1,t}] = \int_0^1 \mathbb{E}_i[\varepsilon_{1,t}|z_{i,t}]\, di. \tag{18}$$

Combining (13) and (14) with the optimal attention from Lemma 1, each agent's conditional expectation is

$$\mathbb{E}_i[\varepsilon_{1,t}|z_{i,t}, s] = \tau_s^* z_{i,t} = \tau_s^*(\varepsilon_{1,t} + \eta_{i,t}). \tag{19}$$

By the law of large numbers, idiosyncratic noise averages out, so that

$$\bar{\mathbb{E}}_t[\varepsilon_{1,t}] = \tau_s^* \varepsilon_{1,t}. \tag{20}$$

The identification failures due to information choice derived below are therefore not caused by noise in expectations, since the noise terms average out across agents. Rather, the failure comes from the structural dependence of attention on the variance of the shock of interest.[7]

3.5 Regime-Dependent Impact Matrix

Substituting (20) into the expectations-augmented transmission equation (9), we recover the standard structural VAR form in (1), with a regime-dependent impact matrix given by

$$\mathbf{B}^{(s)} = \begin{pmatrix} \boldsymbol{\alpha}_1 + \tau_s^* \boldsymbol{\beta}_1, & \mathbf{b}_2, & \cdots & \mathbf{b}_n \end{pmatrix}. \tag{21}$$

Proposition 1 (Regime-Dependent Transmission) *With known variance regimes and endogenous attention, the structural impact matrix is regime-dependent: whenever at least one regime has $\sigma_s^2 > \theta/2$ and $\boldsymbol{\beta}_1 \neq \mathbf{0}$ we have that $\mathbf{B}^{(s)}$ varies with s. This violates Assumption 1 (Structural Invariance), which breaks the usual conditions for identification via heteroskedasticity.*

Proof. By equation (17), $\tau_K^* > \tau_{K-1}^*$ whenever $\sigma_K^2 > \theta/2$. From (21), $\mathbf{b}_1^{(K)} - \mathbf{b}_1^{(K-1)} = (\tau_K^* - \tau_{K-1}^*)\boldsymbol{\beta}_1 \neq \mathbf{0}$ when $\boldsymbol{\beta}_1 \neq \mathbf{0}$. ■

Notably, this result shows that endogenous attention makes the impact matrix $\mathbf{B}^{(s)}$ regime dependent, but says nothing about the lag polynomial $\mathbf{A}(L)$, which remains invariant across regimes. This is a consequence of the assumption in Section 3.1 that the shock of interest is i.i.d., which is standard in SVAR settings. With i.i.d. shocks, the signals and expectations formed in period $t+1$ are independent of those in period t. Attention choices therefore have no effect on the dynamic propagation of the shock, unlike in DSGE models with inattention to persistent shock processes (e.g. Maćkowiak and Wiederholt, 2015).

The assumption of a constant structural impact matrix (Assumption 1), that Proposition 1 shows can fail in the presence of rational inattention, is often simply imposed in practical applications. However, in principle it is possible to test it in some contexts. In Appendix C I therefore discuss if

[7] Equation (20) also implies that including measures of expectations in the VAR will not address the failure of Assumption 1, because the mapping from $\varepsilon_{1,t}$ to $\bar{\mathbb{E}}_t[\varepsilon_{1,t}]$ still varies with the regime.

and how the identification failure outlined here could be detected in these tests, and argue that the tests will often be under-powered in standard macroeconomic applications.

3.6 What Heteroskedasticity Identification Recovers

The econometrician observes reduced-form covariances $\mathbf{\Omega}^{(s)}$ and (incorrectly) assumes a constant B. They then compute the difference:

$$\mathbf{\Omega}^{(H)} - \mathbf{\Omega}^{(L)} = \mathbf{B}^{(H)}\mathbf{\Sigma}_{\varepsilon}^{(H)}(\mathbf{B}^{(H)})' - \mathbf{B}^{(L)}\mathbf{\Sigma}_{\varepsilon}^{(L)}(\mathbf{B}^{(L)})' \tag{22}$$

$$= \sigma_H^2 \left(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1\right)\left(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1\right)' - \sigma_L^2 \left(\boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1\right)\left(\boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1\right)' \tag{23}$$

where H and L denote two regimes such that $\sigma_H^2 > \sigma_L^2$, and we maintain the assumption that only the variance of the first structural shock varies with the regime. All other structural shocks have constant variance and so cancel out in (22).

This is not proportional to $\mathbf{b}_1^{(H)}(\mathbf{b}_1^{(H)})'$ or $\mathbf{b}_1^{(L)}(\mathbf{b}_1^{(L)})'$, outside of the knife-edge case in which $\boldsymbol{\alpha}_1$ and $\boldsymbol{\beta}_1$ are collinear. The identified object is a weighted combination of attention-invariant and attention-sensitive transmission channels, mixed across regimes.

Proposition 2 (Pseudo-Parameter Recovered when Endogenous Attention is Ignored) *Suppose an econometrician applies the standard spectral procedure used when the impact matrix* B *is believed to be constant across regimes: they set* b_1 *proportional to the eigenvector associated with the largest (positive) eigenvalue of* $\Delta\mathbf{\Omega} \equiv \mathbf{\Omega}^{(H)} - \mathbf{\Omega}^{(L)}$.

If $\left(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1\right)'\left(\boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1\right) \neq 0$, *then the resulting estimate* $\hat{b}_1$ *(up to scale) is given by*

$$\hat{b}_1 \;\propto\; \left(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1\right) + k \cdot \left(\boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1\right), \tag{24}$$

where k *is a combination of structural parameters* $\boldsymbol{\alpha}_1, \boldsymbol{\beta}_1$, *variances* σ_H^2, σ_L^2, *and attention* τ_H^*, τ_L^* *defined in Appendix B.2.*

Unless the regime-specific impact vectors are collinear or orthogonal, $\hat{b}_1$ *is generically different from both* $(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1)$ *and* $(\boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1)$. *The resulting impulse responses therefore correspond to a pseudo-parameter rather than the true impact of the shock in either regime.*

Proof. Appendix B.2. ■

That is, an econometrician who ignores endogenous information acquisition will obtain an estimate $\hat{b}_1$, which may appear reasonable, but which does not correspond to the true structural impact vector in either regime. This, in turn, means that impulse responses computed using $\hat{b}_1$ will not reflect the true transmission of the structural shock $\varepsilon_{1,t}$ in either regime.

Corollary 1 (Impulse Responses Implied by Identification Ignoring Endogenous Attention) *Let $\{\mathbf{\Psi}_h\}_{h\geq 0}$ denote the reduced-form moving-average matrices implied by the lag polynomial $A(L)$, so that under regime s the structural impulse response of y_{t+h} to a unit $\varepsilon_{1,t}$ shock is*

$$IRF_s(h) \;\equiv\; \mathbf{\Psi}_h\,(\boldsymbol{\alpha}_1 + \tau_s^*\boldsymbol{\beta}_1), \qquad h = 0, 1, 2, \ldots. \tag{25}$$

If the econometrician identifies $\hat{b}_1$ as in Proposition 2, for every horizon h the estimated impulse response is

$$\widehat{IRF}(h) \;\propto\; \mathbf{\Psi}_h\,\hat{b}_1 \;=\; IRF_H(h) + k\cdot IRF_L(h). \tag{26}$$

Normalizing the estimated impulse response so the impact on variable m at horizon 0 is equal to one (i.e. setting $e_m'\widehat{IRF}(0) = 1$), the estimated IRF is

$$\widehat{IRF}(h) = \frac{IRF_H(h) + k\,IRF_L(h)}{e_m'(IRF_H(0) + k\,IRF_L(0))}. \tag{27}$$

which outside of the knife-edge cases discussed in Proposition 2 differs from both $IRF_H(h)$ and $IRF_L(h)$ for generic h.

Proof. Appendix B.3. ■

Convexity and the sign of k. If the coefficient k is positive, then the pseudo-parameter $\hat{b}_1$ is a convex combination of the true regime-specific impact vectors: the relative weights $\frac{1}{1+k}$ and $\frac{k}{1+k}$ are both in $[0, 1]$. In this case, the naïve estimator is biased relative to either regime, but at least recovers an impact vector between the true regime-specific impact vectors.

However, there is no guarantee in Proposition 2 that k will be positive. In fact, it turns out that k is typically *negative*, implying that $\hat{b}_1$ is a non-convex combination of the true regime-specific impact vectors, and the recovered estimates point outside of the cone between them.

Proposition 3 provides a simple geometric characterization of when k will turn negative.

Proposition 3 (Sign of the Pseudo-Parameter Weight) *The coefficient k in (24) is always real, and satisfies*

$$\operatorname{sign}(k) = -\operatorname{sign}\big(\,(\boldsymbol{\alpha}_1 + \tau_H^*\boldsymbol{\beta}_1)'\,(\boldsymbol{\alpha}_1 + \tau_L^*\boldsymbol{\beta}_1)\,\big). \tag{28}$$

Hence the naïve estimate $\hat{b}_1$ is a non-convex combination of the true regime-specific impact vectors ($k < 0$) if and only if $(\boldsymbol{\alpha}_1 + \tau_H^*\boldsymbol{\beta}_1)'\,(\boldsymbol{\alpha}_1 + \tau_L^*\boldsymbol{\beta}_1) > 0$, i.e. if and only if the two impact vectors form an acute angle; it is a convex combination ($k > 0$) if and only if they form an obtuse angle.

Proof. Appendix B.4 ■

To see why this implies k is typically negative, expand out (28) to give the following condition for $k < 0$:

$$||\boldsymbol{\alpha}_1||^2 + (\tau_H^* + \tau_L^*)(\boldsymbol{\alpha}_1'\boldsymbol{\beta}_1) + \tau_H^*\tau_L^*||\boldsymbol{\beta}_1||^2 > 0. \tag{29}$$

Studying this condition highlights several special cases where non-convexity is guaranteed. First, if the angle between the 'mechanical' and 'expectational' transmission vectors $\boldsymbol{\alpha}_1$ and $\boldsymbol{\beta}_1$ is acute ($\boldsymbol{\alpha}_1'\boldsymbol{\beta}_1 > 0$), the condition is always satisfied and $k < 0$. One particular (and rather strict) example of this is if each element of $\boldsymbol{\alpha}_1$ has the same sign as the corresponding element of $\boldsymbol{\beta}_1$. Economically, that would describe a setting in which attention (weakly) amplifies the impact of the shock on all variables in the system relative to a no-attention baseline.

However, even if $\boldsymbol{\alpha}_1'\boldsymbol{\beta}_1 < 0$, condition (29) remains highly likely to be satisfied for a wide range of parameters. If mechanical transmission channels dominate expectational channels, so $\boldsymbol{\alpha}_1$ is large in magnitude relative to $\boldsymbol{\beta}_1$, the first term in condition (29) will be large relative to the second, and we will obtain $k < 0$. If instead expectational channels come to dominate, that too can cause $k < 0$, as a very large $\boldsymbol{\beta}_1$ implies a large value for the third term, which is also always positive.

That is, for condition (29) to fail, we would require mechanical and expectational impact vectors to point in opposite directions from one another (i.e. be separated by an obtuse angle), *and* for their relative magnitudes to fall in a specific range. As a further example of this, at the largest possible negative magnitude for $\boldsymbol{\alpha}_1'\boldsymbol{\beta}_1$, condition (29) simplifies, such that we obtain convexity (i.e. $k > 0$) if and only if $\tau_L^*||\boldsymbol{\beta}_1|| < ||\boldsymbol{\alpha}_1|| < \tau_H^*||\boldsymbol{\beta}_1||$.[8]

While it is therefore possible to find situations in which the pseudo-parameter is a convex combination of regime-specific vectors, it is rare in practice. Non-convexity is the generic case. In a simple simulation, where I draw many combinations of model parameters, k is negative for more than 99% of parameter draws.[9] Usefully, applying the bias correction methods derived later in this paper yields estimates for the regime-specific impact vectors, which means that condition (28) can be checked. In the monetary policy application in Section 5.3, the sign of k is indeed negative.

This argument is made from a geometric point of view, but there is a more economic interpretation of the non-convexity result. The standard heteroskedasticity estimator compares the estimated impact vectors $\boldsymbol{\alpha}_1 + \tau_H^*\boldsymbol{\beta}_1$ and $\boldsymbol{\alpha}_1 + \tau_L^*\boldsymbol{\beta}_1$. Expecting to find them pointing in the same direction, the eigenvector procedure effectively extracts the difference between these two impacts, and uses that difference for identification. If the true impact vectors for each regime point in slightly *different* directions, as they do with endogenous attention, then the difference between them points in a direction that is outside the cone between them. Therefore, when the regime-specific impact

[8] Formally, the minimum possible $\boldsymbol{\alpha}_1'\boldsymbol{\beta}_1$ is $-||\boldsymbol{\alpha}_1||||\boldsymbol{\beta}_1||$, at which point condition (29) factorizes to $(||\boldsymbol{\alpha}_1|| - \tau_L^*||\boldsymbol{\beta}_1||)(||\boldsymbol{\alpha}_1|| - \tau_H^*||\boldsymbol{\beta}_1||) > 0$, which fails if and only if $\tau_L^*||\boldsymbol{\beta}_1|| < ||\boldsymbol{\alpha}_1|| < \tau_H^*||\boldsymbol{\beta}_1||$.

[9] The simulation draws parameters, for VARs with $2-6$ variables, from the distributions: elements of $\boldsymbol{\alpha}_1$, $\boldsymbol{\beta}_1$ from $N(0,1)$; $\theta \sim U(0,2)$; $\sigma_L^2 \sim U(0.2, 2.2)$; and $\sigma_H^2 = \sigma_L^2 + U(0,4)$ to ensure $\sigma_H^2 \geq \sigma_L^2$. I find $k < 0$ in 99.2% of the 2 million simulations. Within the set with $\boldsymbol{\alpha}_1'\boldsymbol{\beta}_1 < 0$ (50% of draws by construction), $k < 0$ in 98.4% of simulations.

vectors point in broadly similar directions (i.e. when the angle between them is acute), the standard estimator recovers a non-convex combination of them. The convex case only arises if attention reorients transmission so violently that the impact vectors end up pointing in opposite directions.[10]

Finally, note that even if $\hat{b}_1$ is a convex combination of $\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1$ and $\boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1$, this may not be enough to guarantee that normalized IRFs are a convex combination of true regime-specific IRFs, because the normalization step introduces further non-linearity. Formally, define $N_s \equiv e_m' IRF_s(0)$ as the impact of the shock on the normalizing variable m in regime s, and define $\widetilde{IRF}_s(h) \equiv IRF_s(h)/N_s$ as the corresponding normalized true regime-specific response. Then (27) becomes

$$\widehat{IRF}(h) = \frac{N_H}{N_H + kN_L} \widetilde{IRF}_H(h) + \frac{kN_L}{N_H + kN_L} \widetilde{IRF}_L(h), \tag{30}$$

where the two weights sum to one.

That is, the normalized IRF obtained by a standard identification via heteroskedasticity is a convex combination of true regime-specific IRFs if $k > 0$ *and* if N_H and N_L share the same sign. If, on impact, the normalizing variable is one which the shock affects in different directions across regimes, that creates a further non-convexity. This particular concern is not relevant in the monetary policy application below (see footnote 23), but may appear in other settings.

When is the bias negligible? The discussion above concerns whether the bias pushes the estimated coefficients and IRFs outside of the range of the true regime-specific objects. It does not, however, say anything about the magnitude of the bias. Since this may be of particular interest to practitioners, I discuss it explicitly here.

The bias characterized in Proposition 2 and Corollary 1 arises from the interaction of two forces: agents adjust their attention across variance regimes, and this attention affects shock transmission. If either channel is absent, identification via heteroskedasticity remains valid. This, however, involves strong structural economic assumptions that are unlikely to hold in many contexts.

In general, attention may be constant across regimes for three reasons. First, if attention costs are very large relative to the payoffs of being informed, attention will be 0 in all regimes. In the model above, this occurs if $\theta > 2\sigma_s^2$ for all regimes s. Second, if attention costs are very small, then attention approaches 1 in all regimes.[11] The findings of time-varying attention across various contexts and agents suggest that these cases are rare (e.g. Mondria and Quintana-Domeque, 2013; Song and Stern, 2024; Flynn and Sastry, 2024; Macaulay, 2025).

Third, attention may be constant across regimes if agents do not know that the regime has

[10] Interestingly, this means that non-convexity is more likely if the setup appears more standard. That is, if most of a shock's transmission is mechanical, and expectations-driven transmission is small, then it is even more likely that $k < 0$. Note however that non-convexity is not sufficient to imply a large bias, hence the magnitude discussion below.

[11] Formally, the difference in attention between two regimes, when both regimes lead to the interior solution for attention, is $\tau_H^* - \tau_L^* = \theta/2(\sigma_L^{-2} - \sigma_H^{-2})$, which approaches 0 as $\theta \to 0$.

changed (i.e. if Assumption 3 does not hold). Rational inattention models typically assume that agents know the distribution of shocks they are facing,[12] but if this is not the case then identification via heteroskedasticity would not be subject to the bias characterized above, as long as the econometrician is still able to identify the regimes ex post. For the monetary policy application in Section 5 this is unlikely to hold, as the high-variance regime consists of scheduled FOMC announcement days. There is also a substantial empirical literature documenting attention responding to volatility in a range of other contexts (e.g. Andrei and Hasler, 2015; Andrei et al., 2023; Mikosch et al., 2024; Benchimol et al., 2025). Importantly, if the researcher has access to a proxy for attention to the relevant variable(s), the variation of attention across regimes can easily be tested, as I do in Section 5.2 below.

Another situation in which this bias from information acquisition will be negligible is if expectations are themselves a negligible part of shock transmission. Formally, if $\boldsymbol{\beta}_1 = 0$, then equation (24) implies $\hat{b}_1 \propto \boldsymbol{\alpha}_1$, as in standard arguments for identification via heteroskedasticity. However, a vast literature, from Lucas (1972) (and earlier influences) onwards, has shown that expectations are important in a very wide range of macroeconomic and financial dynamics, so such instances where every variable in a VAR is independent of attention are likely to be rare.[13]

4 Bias Correction

The variation in the impact matrix across regimes highlighted above depends on the attention of relevant agents in the economy. In this section I propose two ways of correcting this bias.

4.1 Correction Method 1: Using an Attention Proxy

Geometric preliminaries. The observed difference between two reduced-form covariance matrices across regimes ($\Delta\boldsymbol{\Omega}_{H,L} \equiv \boldsymbol{\Omega}^{(H)} - \boldsymbol{\Omega}^{(L)}$) is defined in equation (23). If Assumption 1 held, this difference matrix would be rank 1, with one non-zero (positive) eigenvalue (Rigobon, 2003). However, with endogenous attention, Assumption 1 fails. Since $\Delta\boldsymbol{\Omega}_{H,L}$ is then the difference of two rank-1 positive semi-definite matrices, it will be rank 2, with one positive and one negative eigenvalue.

The eigen-decomposition of $\Delta\boldsymbol{\Omega}_{H,L}$ can be written as

$$\Delta\boldsymbol{\Omega}_{H,L} = \lambda_1 u_1 u_1' + \lambda_2 u_2 u_2' \tag{31}$$

[12] See Ellison and Macaulay (2021) for an example where this is not the case.

[13] A final knife-edge case is if $\boldsymbol{\alpha}_1$ and $\boldsymbol{\beta}_1$ are collinear (i.e. $\boldsymbol{\alpha}_1 = \kappa\boldsymbol{\beta}_1$ for some scalar κ). The impact vector then varies in magnitude across regimes but not in direction. The covariance difference matrix remains rank one, and standard heteroskedasticity-based identification recovers the correct impact vector direction. This case may be unlikely to hold exactly in practice but illustrates that the bias stems from rotation of the impact vector, not merely rescaling.

where $\lambda_1 > 0 > \lambda_2$ are the eigenvalues, and u_1, u_2 are the corresponding eigenvectors. The true scaled impact vectors, which I denote $w_H \equiv \sigma_H(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1)$ and $w_L \equiv \sigma_L(\boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1)$, must lie in the plane spanned by u_1 and u_2.

With this, we can therefore characterize the set of all candidate impact vectors that are consistent with the observed covariance difference. The solution to $w_H w_H' - w_L w_L' = \lambda_1 u_1 u_1' + \lambda_2 u_2 u_2'$ is given by a hyperbolic rotation of the eigenvectors:

$$w_H(\phi) = \sqrt{\lambda_1}\cosh(\phi)u_1 + \sqrt{|\lambda_2|}\sinh(\phi)u_2 \tag{32}$$

$$w_L(\phi) = \sqrt{\lambda_1}\sinh(\phi)u_1 + \sqrt{|\lambda_2|}\cosh(\phi)u_2 \tag{33}$$

where $\phi \in \mathbb{R}$ is an unknown rotation parameter. Identification of the true IRFs reduces to pinning down the scalar ϕ.

Using an attention proxy. Provided that attention is interior in both regimes, we can use Lemma 1 to write the ratio of structural shock variances across regimes as

$$r \equiv \frac{\sigma_H^2}{\sigma_L^2} = \frac{1 - \tau_L^*}{1 - \tau_H^*}. \tag{34}$$

If we observe the relative levels of (in)attention, we can therefore infer the relative structural shock variances, which are not typically observed in Rigobon (2003)-style exercises.

With this in hand, we can now solve for ϕ and thus the bias-corrected IRFs, if we impose two further restrictions on the structure of transmission.

Assumption 4 (Attention-Invariant Variable) *The effect of the shock of interest on one of the variables in the system on impact is unaffected by attention: $\beta_{1,k} = 0$ for one specified variable k.*

Assumption 5 (Known Attention Direction) *The effect of attention on the impact effect of the shock of interest can be signed for one of the variables of the system: we can specify either $\beta_{1,j} > 0$ or $\beta_{1,j} < 0$ for one variable j.*

Assumption 4 requires that there is at least one variable in the system for which the transmission on impact is purely mechanical, and does not depend on expectations, thus $\beta_{1,k} = 0$. The dynamic effects can still depend on attention through the impact effects on other variables, which then feed into variable k over time through lags of y_t in the VAR. In monetary policy applications, this attention-invariant variable could plausibly be the policy rate, which moves due to the policymaker actions whether or not people pay attention to them. We use this restriction to identify the rotation parameter ϕ in equations (32) and (33), and discuss it further below.

In general, there will be two roots to the resulting equation, yielding two possible values for ϕ. Assumption 5 is then employed to select between them. If economic theory suggests that a particular variable should respond more strongly on impact to a shock when agents are paying more attention, this provides the necessary restriction. As we are selecting between only two possible roots, we do not need to form a prior on the magnitude of this attention effect.

With these two assumptions, we proceed in the following way. If $\beta_{1,k} = 0$, then we have that the impact of the shock on that variable is invariant across regimes:

$$\frac{w_{H,k}(\phi)}{\sigma_H} = \frac{w_{L,k}(\phi)}{\sigma_L} \implies w_{H,k}(\phi) = \sqrt{r} w_{L,k}(\phi) \tag{35}$$

Substituting equations (32) and (33) into (35) and rearranging gives:

$$\tanh(\phi^*) = \frac{\sqrt{\lambda_1} u_{1,k} - \sqrt{r|\lambda_2|} u_{2,k}}{\sqrt{r\lambda_1} u_{1,k} - \sqrt{|\lambda_2|} u_{2,k}}. \tag{36}$$

This yields two potential roots for ϕ, corresponding to the sign ambiguity of the eigenvectors. To ensure a unique and economically meaningful solution, we normalize the recovered vectors such that the impact on a chosen anchor variable has a given sign, and then use Assumption 5 to select between the roots of ϕ^*.

These restrictions, and equation (36), therefore allow us to recover ϕ^*, and thus $w_s(\phi)$. With an appropriate scale normalization,[14] we then obtain the impact vector $\boldsymbol{\alpha}_1 + \tau_s^* \boldsymbol{\beta}_1$. From these we can compute bias-corrected impulse response functions for both regimes.

Geometric intuition. The impact vectors in different regimes differ only through the scalar attention parameter τ_s^*, so regime variation in the impact vector is one-dimensional. This is why the covariance difference matrix $\Delta\Omega_{H,L}$ has rank two, and so its eigendecomposition identifies the two-dimensional plane in $\mathbb{R}^n$ in which the scaled impact vectors w_H and w_L must lie. While this plane is observable, the location of the impact vectors within it is not: there is a one-parameter family of candidate pairs (w_H, w_L) consistent with the data, indexed by the rotation ϕ.

The identifying restrictions detailed above progressively narrow this family. The attention proxy determines the variance ratio r, which defines two hyperbolas in the space of the unscaled impact vectors $v_s = w_s/\sigma_s$: one traced by the candidate v_H and the other by the candidate v_L. Assumption 4 then requires that v_H and v_L agree on one coordinate, which reduces the candidate solutions to two (v_H, v_L) pairs. Finally, Assumption 5 selects between them.

[14] Such normalizations are standard in SVAR analysis (see e.g. Section 2). In the application in Section 5.3 the normalization is that the shock moves the 2-year treasury yield by 25bps on impact.

Sensitivity. If Assumptions 4 and 5 hold exactly and r is measured without error, this method exactly corrects for the bias derived in Proposition 2 in population. In addition, Appendix D presents a simulation that shows the correction recovers the true impulse responses with little error in finite samples of the size used in Section 5.

In practice, however, both Assumption 4 and the assumption that we can observe a perfect attention proxy may be difficult to satisfy precisely. In simple monetary policy contexts, as mentioned above, the policy rate might plausibly be attention-invariant on impact as required, but outside of this case such a variable may be harder to find. Similarly, available measures of attention are known to be noisy in many contexts, which may lead to inaccuracy in the calibrated ratio r.

For this reason, in Appendix D I also test the sensitivity of the bias correction to violations of Assumption 4 and errors in the calibration of r. I simulate datasets, varying the true impact effect of attention for the variable assumed invariant ($\beta_{1,k}$) or the true variance ratio r. In each simulation, I estimate a VAR and conduct the bias correction using the (now incorrect) baseline assumptions. The results indicate that the correction is robust to moderate errors in $\beta_{1,k}$ and r. When $|\beta_{1,k}|$ is within 10% of the attention effect on other variables in the system, or when r is mismeasured by 10%, the method still delivers more than 90% of the improvement over the uncorrected estimator that would arise under exact assumptions. The method therefore remains valuable even when the identifying assumptions are not exact.

4.2 Correction Method 2: Multiple Regimes

While Method 1 provides a closed-form solution, it relies on the existence of an attention-invariant variable (Assumption 4), and of a reliable proxy for attention. In applications where rational inattention permeates all variables in the system (i.e., β_1 has no zero elements), or where attention-related data is unavailable, this will not be appropriate. I show here that observing more than two variance regimes allows for an alternative bias correction, which exploits the geometric constraints implied by the attention mechanism.

Specifically, consider the case with three variance regimes, $s \in \{L, M, H\}$. The structural impact vector in regime s is:

$$v_s = \alpha_1 + \tau_s^* \beta_1 \tag{37}$$

Geometrically, this equation implies that the impact vectors $\{v_L, v_M, v_H\}$ must be *collinear*. They lie on a single line in $\mathbb{R}^n$ with intercept α_1 and direction β_1. This collinearity is a strong restriction which allows us to estimate the structural parameters for each regime.

The estimator. First, normalize the parameters specific to the low-variance regime to $\sigma_L^2 = 1$ and $\tau_L^* = 0$, effectively interpreting α_1 as the baseline transmission and τ_s^* as the *relative* increase in attention in the medium- and high-variance regimes. We can then define the vector of remaining deep structural parameters as $\Theta = [\alpha_1', \beta_1', \tau_M^*, \tau_H^*, \sigma_M^2, \sigma_H^2]'$.

The theoretical difference between the covariance matrix in regime s and the baseline regime L is:

$$\Delta\Omega_{s,L}(\Theta) = \sigma_s^2(\alpha_1 + \tau_s^*\beta_1)(\alpha_1 + \tau_s^*\beta_1)' - \alpha_1\alpha_1' \tag{38}$$

Our aim is to estimate Θ. If there are n variables in $\mathbf{y}_t$ (i.e. if $\Delta\Omega_{s,L}$ is $n \times n$), this contains $2n+4$ parameters: n elements of α_1, n elements of β_1, plus τ_M^*, τ_H^*, σ_M^2, σ_H^2.

We estimate Θ by minimizing the distance between these model-implied moments and the data:

$$\hat{\Theta} = \arg\min_{\Theta} \sum_{s \in \{M,H\}} \left\| \text{vec}(\Delta\Omega_{sL}^{obs}) - \text{vec}(\Delta\Omega_{sL}(\Theta)) \right\|_W^2 \tag{39}$$

where W is a positive definite weighting matrix.

Identification conditions and effective degrees of freedom. A crucial feature of the model is that while variance changes are driven by a single structural shock, the resulting covariance difference matrices $\Delta\Omega_{s,L}$ are of rank two. This structural constraint implies that the elements of $\Delta\Omega_{s,L}$ are not independent; for $n \geq 3$, the matrix contains only $2n-1$ independent pieces of information rather than the usual $n(n+1)/2$. Moreover, $\Delta\Omega_{M,L}$ and $\Delta\Omega_{H,L}$ are not independent of each other, as they share a common column space ($span\{\alpha_1, \beta_1\}$).

Accounting for these constraints, the pair $(\Delta\Omega_{M,L}, \Delta\Omega_{H,L})$ contains $2n+2$ independent degrees of freedom.[15] Since Θ contains $2n+4$ free parameters to be estimated, we require two extra restrictions or normalizations.

This under-identification occurs because of two scale indeterminacies. First, doubling β_1 and halving both τ_M^* and τ_H^* does not affect the resulting $\Delta\Omega_{s,L}$. To address this I therefore normalize $\tau_H^* = 1$. Second, the model is similarly invariant to doubling $v_s = \alpha_1 + \tau_s^*\beta_1$ and halving σ_s. For this, I set $v_{H,k} = 1$ for a chosen anchor variable k, which fixes the scale of the impact vector in the high-variance regime. This is analogous to the standard scale normalization required in heteroskedasticity-based identification (discussed in Section 2), expressed here in terms of the structural parameters. Appendix D shows that this method is able to recover impulse responses close to the true regime-dependent responses in a finite-sample simulation.

Incorporating external information. In principle, the researcher could combine the approach above with proxy measures of attention, as used in Method 1 above. Restrictions on τ_H^*/τ_M^*, for example, reduce the dimensionality of the estimation problem, thus making it possible to estimate

[15]The shared two-dimensional column space contributes $2(n-2)$ degrees of freedom, and the representation of each matrix within that subspace contributes 3, giving $2(n-2) + 2 \times 3 = 2n+2$ in total.

Θ with one fewer normalization. Even maintaining the normalizations, such external information may still be helpful in making the estimates of α_1 and β_1 more precise.

5 Empirical Evidence: Attention to Monetary Policy

I now apply the insights developed above to the analysis of unconventional monetary policy shocks at the zero lower bound (ZLB).[16] Specifically, I follow Wright (2012) in using daily data on six asset prices to estimate the effects of unconventional monetary policy shocks at the zero lower bound, assuming that monetary policy shocks have a greater variance on FOMC announcement days than non-announcement days.[17] I extend the Wright (2012) sample to the full ZLB period (November 3 2008-December 9 2015). The variables are: the 10-year and 2-year zero-coupon Treasury yields, 5-year and 5-to-10-year forward TIPS breakeven inflation rates, and Moody's AAA and BAA corporate bond yields.[18]

I start with two tests of the rational inattention model of Section 3, concerning the eigenvalue structure of the reduced-form covariance difference matrix $\Delta\boldsymbol{\Omega}_{H,L}$ and an attention proxy based on Google Trends search data. I then apply the bias-correction method developed in Section 4.1, and find substantial differences to uncorrected results in the estimated effects of monetary policy, especially on expected inflation.

5.1 Eigenvalue Check

Under the standard identifying assumption (Assumption 1), the difference in reduced-form covariance matrices between announcement and non-announcement days $\Delta\boldsymbol{\Omega}_{H,L} \equiv \boldsymbol{\Omega}^{(H)} - \boldsymbol{\Omega}^{(L)}$ is a rank-one positive semi-definite matrix with exactly one positive eigenvalue. In contrast, under the rational inattention model developed in Section 3, this difference matrix is rank two, with one positive and one negative eigenvalue.

I test this prediction using the data outlined above. Following the specification in Wright (2012), I estimate a VAR(1) on the six variables and compute the covariance matrices of the residuals $\boldsymbol{\Omega}^{(H)}$ and $\boldsymbol{\Omega}^{(L)}$. The one change I make is that I exclude the data from December 1 2008, when Ben Bernanke made a speech indicating that the Federal Reserve would begin buying Treasuries. On

[16] Wright (2012) also conducts a high-frequency event study, a common combination since under Assumption 1 event study regressions coincide with SVARs identified via heteroskedasticity in the limit as the variance of non-policy shocks goes to 0. That equivalence no longer holds with endogenous attention. Appendix A.3 contains a detailed discussion.

[17] Wright (2012) also includes the days of five landmark policy speeches in the high-variance regime. In the absence of an objective criterion for which speeches should count in the later part of the sample (after Wright (2012) was written), I restrict myself to scheduled FOMC announcement days only.

[18] The data are obtained from the Federal Reserve Bank of St. Louis replication archive, accessible at https://fredaccount.stlouisfed.org/public/datalist/1111. The sample period contains 56 announcement days.

Table 1: Eigenvalues of the Covariance Difference Matrix

	(1) Point Estimate	(2) Bootstrap 95% CI	(3) Point Estimate	(4) Bootstrap 95% CI
λ_1 (smallest)	-0.0027	$[-0.0123, -0.0010]$	-0.0058	$[-0.0294, -0.0029]$
λ_2	-0.0013	$[-0.0026, -0.0001]$	-0.0033	$[-0.0060, -0.0005]$
λ_3	-0.0001	$[-0.0005, 0.0001]$	-0.0002	$[-0.0013, -0.0000]$
λ_4	0.0001	$[-0.0001, 0.0007]$	0.0000	$[-0.0002, 0.0016]$
λ_5	0.0009	$[0.0001, 0.0036]$	0.0018	$[-0.0000, 0.0091]$
λ_6 (largest)	0.0099	$[0.0023, 0.0224]$	0.0177	$[0.0018, 0.0430]$
Sample period	ZLB	ZLB	Wright	Wright

Notes: Eigenvalues of $\Delta\mathbf{\Omega}_{H,L} = \mathbf{\Omega}^{(H)} - \mathbf{\Omega}^{(L)}$ from VAR(1) residuals. Bootstrap confidence intervals based on 5,000 replications. All samples exclude December 1 2008 (see text).

this day the 5-year TIPS breakeven inflation rate jumped up by 192 basis points, which is 21 times larger than the standard deviation of daily changes over the sample period. If not excluded, this one outlier observation substantially inflates the variance of breakeven inflation on non-announcement dates, giving rise to concerns about weak identification (Lewis, 2022).

Table 1 reports the eigenvalues of the covariance difference matrix $\Delta\mathbf{\Omega}_{H,L}$. Columns 1 and 2 show the results for the full ZLB sample. The matrix has three negative eigenvalues, with the smallest equal to -0.0027. Under constant transmission (Assumption 1), all eigenvalues should be non-negative with at most one strictly positive. The presence of negative eigenvalues suggests a violation of this constant-transmission assumption. For comparison, columns 3 and 4 show the results from the restricted subsample used in Wright (2012), which show similar patterns.

To assess statistical significance, I construct bootstrap confidence intervals by resampling announcement and non-announcement days separately with replacement. The 95% confidence interval for the minimum eigenvalue in the full ZLB sample is $[-0.012, -0.001]$, which lies entirely below zero. This suggests that the rank-one assumption is unlikely to hold. Moreover, 100% of bootstrap replications produce at least one negative eigenvalue, indicating that this finding is robust to sampling uncertainty.[19]

The two eigenvalues of largest magnitude (0.010 and -0.003) together account for 84% of the total sum of absolute eigenvalues. This concentration is consistent with an approximately rank-two structure, as predicted by the rational inattention model. Eigenvalues λ_3–λ_5 are an order of magnitude smaller and likely reflect estimation noise. λ_2 is less clearly near-zero, possibly

[19] Wright (2012) conducts a GMM test of the null that $\Delta\mathbf{\Omega}_{H,L}$ is rank 1 against the unrestricted alternative that it cannot be represented by a single rank-1 factor. Two features make this less informative here than Table 1. First, such rank tests are known to be low-powered at the sample sizes typical in macroeconomics (Appendix C), so a failure to reject rank 1 is weak evidence against rank 2, particularly when (as here) the second eigenvalue is small in magnitude. Second, the test pools all departures from rank 1 together, and so cannot isolate the specific signature of the inattention mechanism: a *negative* second eigenvalue. Consistent with Wright (2012), the test does not reject rank 1 ($p = 0.22$ over the full sample, $p = 0.12$ in the Wright subsample), even though the eigenvalue estimates point to a rank-2 structure.

suggesting some other failure of Assumption 1 beyond the optimal information choice studied here.

The finding of negative eigenvalues in $\Delta\mathbf{\Omega}_{H,L}$ is consistent with the mechanism proposed in this paper. However, it should be noted that these negative eigenvalues could also arise if the variances of multiple structural shocks, not just the monetary policy shock, change between announcement and non-announcement days. In particular, the single policy shock identified here aggregates potentially distinct forward guidance, large-scale asset purchase, and Fed information shocks; if the relative variances of these components differ between regimes, $\Delta\mathbf{\Omega}_{H,L}$ would have rank greater than one even under fully attention-invariant transmission. The eigenvalue analysis here is therefore suggestive, but is not sufficient to single out the rational inattention mechanism as the explanation. The next subsection provides complementary evidence directly documenting that attention to monetary policy spikes on FOMC announcement days.

5.2 Attention to Monetary Policy Between Regimes

I now test if decision-makers pay more attention to monetary policy on FOMC announcement days than on non-announcement days, using Google Trends search volume as a proxy for attention, as in e.g. Da et al. (2011, 2015). Specifically, I use the `trendecon` procedure and package from Eichenauer et al. (2022) to construct a daily series of Google searches for the term "Federal Reserve" from January 1 2006 to June 30 2019.[20] Since Google Trends does not provide daily data over this length of sample, the index is constructed by splicing together multiple Google Trends queries, with adjustments to ensure consistency of units across windows (see Eichenauer et al. (2022) for details of the methodology, and Wang (2025) for a similar procedure). This means that the resulting index is not exactly scaled to be within $[0, 100]$ as is typical for stand-alone Google Trends queries. The interpretation, however, is similar, as 98.8% of the observations lie in this range.

Table 2 shows the average index values on announcement and non-announcement days for the ZLB sample that is the main sample of interest, as well as the shorter sample analyzed by Wright (2012), and the full sample that includes non-ZLB periods. The patterns are similar across samples, with search activity approximately two-thirds higher on announcement days. Even in the relatively short Wright (2012) subsample, these differences are strongly significant.

While indicative, the greater search volume on announcement days is not itself enough to reject the assumption of invariant transmission (Assumption 1). If attention rises only with shock

[20] Since this series requires multiple calls to the Google Trends API, it can only cover a single keyword at a time, and cannot sum multiple keyword search intensities. Constructing an index from multiple `trendecon` queries would not produce an interpretable result, as each search term would have its own scale. However, from the monthly data available directly from Google Trends, I find that 'Federal Reserve' has more than $10\times$ the search activity of 'FOMC,' and more than $20\times$ the activity of 'Federal Funds Rate' over the 2006-2025 period. As a robustness check, in Appendix E I re-run the analysis of Table 2 for the search term 'Monetary Policy,' which has approximately 15% of the search activity of 'Federal Reserve.' All results are robust.

Table 2: Attention to the Federal Reserve: FOMC vs. Non-FOMC Days

	Observations		Mean Search Index			
Sample	FOMC	Non-FOMC	FOMC	Non-FOMC	Difference	t-statistic
Wright (2008–2011)	23	1,039	86.106	49.751	36.356	9.011
ZLB (2008–2015)	56	2,537	86.297	50.132	36.164	14.141
Full (2006–2019)	113	4,816	88.975	46.427	42.547	22.882

Notes: Google Trends search index for "Federal Reserve" constructed using the Eichenauer et al. (2022) methodology. FOMC days are scheduled Federal Open Market Committee announcement days. The t-statistic tests the null hypothesis that mean attention is equal across FOMC and non-FOMC days.

realizations (not variances), then we would see greater attention on announcement days because the average shock magnitude is greater on those days, but there would be no difference in the impact of a unit shock in each regime. To test if this is the case, I therefore estimate

$$\log(A_t) = \alpha + \beta\,\mathrm{FOMC}_t + \gamma'\,|S_t| + \delta' X_t + \varepsilon_t, \tag{40}$$

where A_t is the daily search index (trimmed to drop the smallest and largest 1% of observations), $FOMC_t$ is an indicator equal to 1 on FOMC meeting days, and $|S_t|$ are measures of realized monetary policy shocks.[21] Specifically, I use the absolute values of the Federal Funds Rate, Forward Guidance, and LSAP factors from the high-frequency identification in Swanson (2021). X_t includes controls for day-of-week effects, year-month fixed effects, and two lags of the dependent variable.

The rational inattention mechanism studied above implies that attention to monetary policy should be higher on announcement days irrespective of the realized shocks, i.e. $\beta > 0$. If instead the increased searches on announcement days are entirely due to shock realizations, not the volatility regime, then $\beta = 0$ and all differences in attention between regimes are captured by $\gamma > 0$. Since there are only a small number of announcements in the period analyzed by Wright (2012), I estimate this regression only for the ZLB sample (2008-2015) and the full sample (2006-2019).

The results are presented in Table 3. The first column presents the model omitting FOMC_t and $|S_t|$. Column 2 adds FOMC_t, and column 3 adds $|S_t|$. As predicted by the rational inattention model, β is large and significantly greater than 0 across specifications where FOMC_t is included. The γ coefficients on absolute monetary policy shocks are typically close to 0 and not significant, with the exception of LSAP shocks in the ZLB period. Indeed, it is reassuring that this element of γ is positive and significant, as it provides a sense-check on the attention measure: during the ZLB, when asset purchases were a large and unprecedented feature of US monetary policy, we should expect more attention to the Federal Reserve after large LSAP shocks.

[21] The sample trimming is completed separately for each sample period, so the ZLB-period regressions are run for the days in which A_t is in the middle 98% of the observations for that period. The trimming removes the small number of negative values, so using $\log(A_t)$ does not involve any further sample restrictions.

A further check that attention is determined largely by the regime rather than the shock realization comes from studying the R^2 of these regressions. Adding shock size in columns 3 and 6 only adds a minimal amount to the R^2, an order of magnitude less than is added by the inclusion of FOMC_t in columns 2 and 5. This adds further support to the conclusion that the majority of differences in search volumes between regimes are not driven by the size of the shock realization.

Table 3: Attention to the Federal Reserve

	(1)	(2)	(3)	(4)	(5)	(6)
FOMC day		0.223***	0.163**		0.319***	0.271***
		(0.025)	(0.074)		(0.026)	(0.052)
Fed funds rate shock (abs.)			-0.217			-0.051
			(0.439)			(0.263)
Forward guidance shock (abs.)			0.044			0.091
			(0.041)			(0.057)
LSAP shock (abs.)			0.186***			0.023
			(0.054)			(0.079)
Observations	2,542	2,542	2,542	4,822	4,822	4,822
R-squared	0.7869	0.7909	0.7913	0.7921	0.7986	0.7987
Sample period	ZLB	ZLB	ZLB	Full	Full	Full

Day-of-week and year-month fixed effects, and two lags of log(attention), included in all specifications.
Robust standard errors in parentheses. $*p < 0.10, **p < 0.05, ***p < 0.01$

5.3 Bias-Corrected IRFs

I now apply the bias-correction of Section 4.1 to the structural VAR specified as in Wright (2012).[22] As described above, this requires two extra assumptions beyond those currently made, and an observation of the relative (in)attention between variance regimes. I choose these as follows.

Assumption 4. The impact effect of monetary policy shocks on the 2-year Treasury yield is unaffected by attention.[23]

[22] Correction method 2 described in Section 4.2 is not applicable here as we only observe two variance regimes. For robustness checks Wright (2012) further divides his high-variance regime into "especially important" days and less important days, but this is based on the realized policy changes on those days, not on the volatility that was known ex ante to market participants and other economic agents. This extra division would not therefore satisfy Assumption 3 that is required for the model developed in this paper to be applicable. I provide an application demonstrating correction method 2, to the effects of labor market shocks on asset prices, in Appendix F.

[23] Since this will also be our normalizing variable, the non-convexity channel through impact vectors of different signs highlighted in equation (30) will not apply here. However, the non-convexity from $k < 0$ highlighted in Proposition 3 does apply, as we will estimate that the angle between $(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1)$ and $(\boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1)$ has cosine $= 0.85 > 0$, so $k < 0$ by condition (28).

Assumption 5. The impact effect of monetary policy shocks on the BAA corporate bond yield is larger in magnitude in the high-variance/high-attention regime.

Attention measure. In the Google Trends series above, the mean of the search activity index is 86.3 in the high-variance regime, and 50.1 in the low-variance regime. Taking an index of 0 to reflect no attention ($\tau_{i,t} = 0$) and 100 to reflect complete attention ($\tau_{i,t} = 1$), and assuming a linear relationship between the search index and $\tau_{i,t}$, we obtain $\frac{1-\tau_L^*}{1-\tau_H^*} = \frac{1-0.501}{1-0.863} = 3.64$.

Discussion. The ideal choice of attention-invariant variable for Assumption 4 in monetary policy settings is the policy rate, which moves mechanically with policy actions. However, since my purpose is to identify the effects of unconventional monetary policy shocks at the zero lower bound, the sample is chosen such that the target federal funds rate is constant at the lower bound (0.25%) in all periods, and so is not included in the VAR. However, prior literature provides a mechanism that suggests the 2-year Treasury yield is a plausible alternative.

Wright (2012) argues that the unconventional monetary policy shocks in the sample operate largely through the behavior of preferred-habitat investors and sophisticated arbitrageurs, as in models such as Vayanos and Vila (2021). This mechanism works through portfolio rebalancing by institutional investors with mandates tied to specific maturities, and arbitrageur responses to resulting price pressure. Since these portfolio adjustments are largely mechanical responses to changes in relative bond supplies, they do not require investors to form expectations about the macroeconomic implications of policy.

This mechanism is likely to be especially dominant for shorter maturity assets. Assets with longer maturities may also be affected by the impact of policy announcements on expectations, both of future macroeconomic developments (Boneva et al., 2016; Melosi, 2017) and monetary policy (Bauer and Rudebusch, 2014; Bhattarai et al., 2023).[24] For this reason I impose attention invariance on the 2-year Treasury yield rather than the 10-year Treasury yield.

Despite these arguments, Assumption 4 remains rather strong ex ante. I provide ex post validation at the end of this subsection, along with a robustness test relaxing the assumption. In addition, recall that the simulation results in Appendix D show that the correction remains effective even when Assumption 4 is moderately violated.

For Assumption 5, I consider the impact of monetary policy shocks on corporate bond yields. Part of this transmission is likely to occur through monetary policy's effect on credit spreads: when investors pay closer attention to monetary policy announcements, they update more strongly on what the announcement signals about the Federal Reserve's future policy plans. This would lead them to

[24] Wright (2012) emphasizes the latter channel, noting that "*LSAPs could also work in other ways, such as by affecting agents' expectations of the future course of monetary policy*"(p. F447).

extract more information about future corporate default probabilities than those who observe policy changes only passively through their effect on Treasury yields. This expectational channel should be most pronounced for higher-risk bonds, so I impose the assumption on the BAA yield.

The key threat to this assumption is if 'Delphic forward guidance' (as defined in Campbell et al., 2012) was particularly powerful during the sample period. In that case, monetary policy shocks that lower Treasury yields could have caused attentive investors to negatively revise their expectations of the economy's performance, and thus to demand higher credit spreads. Greater attention would therefore offset the mechanical decline in yields driven by the lower Treasury yields. In other words, I am assuming here that 'Odyssean' forward guidance over future interest rates was more important than these Delphic effects in the ZLB period. This is consistent with the prominent role for explicit interest rate guidance in the ZLB period (see e.g. the Jackson Hole comments of Woodford, 2012), with the substantially larger share of Treasury yield variation attributed to Odyssean than to Delphic shocks in Jarociński (2024) (his Appendix Table D.2), and indeed with recent literature suggesting that part of the apparent information effects of monetary policy shocks may be due to other non-guidance factors (Bauer and Swanson, 2023a,b; Swanson et al., 2025).

In addition, note that Assumption 5 is only used to select between the two roots of equation (36). As such, a range of different possible assumptions would deliver identical impulse response estimates. For example, Figure 1 below shows that assuming attention amplifies the impact of monetary policy shocks on AAA corporate bond yields, rather than BAA yields, would select exactly the same root for ϕ^*, and thus would imply exactly the same impulse responses. Indeed, while the impact response of the AAA yield is also amplified by attention, the amplification is smaller than in the BAA yield, consistent with the default-risk mechanism outlined above.

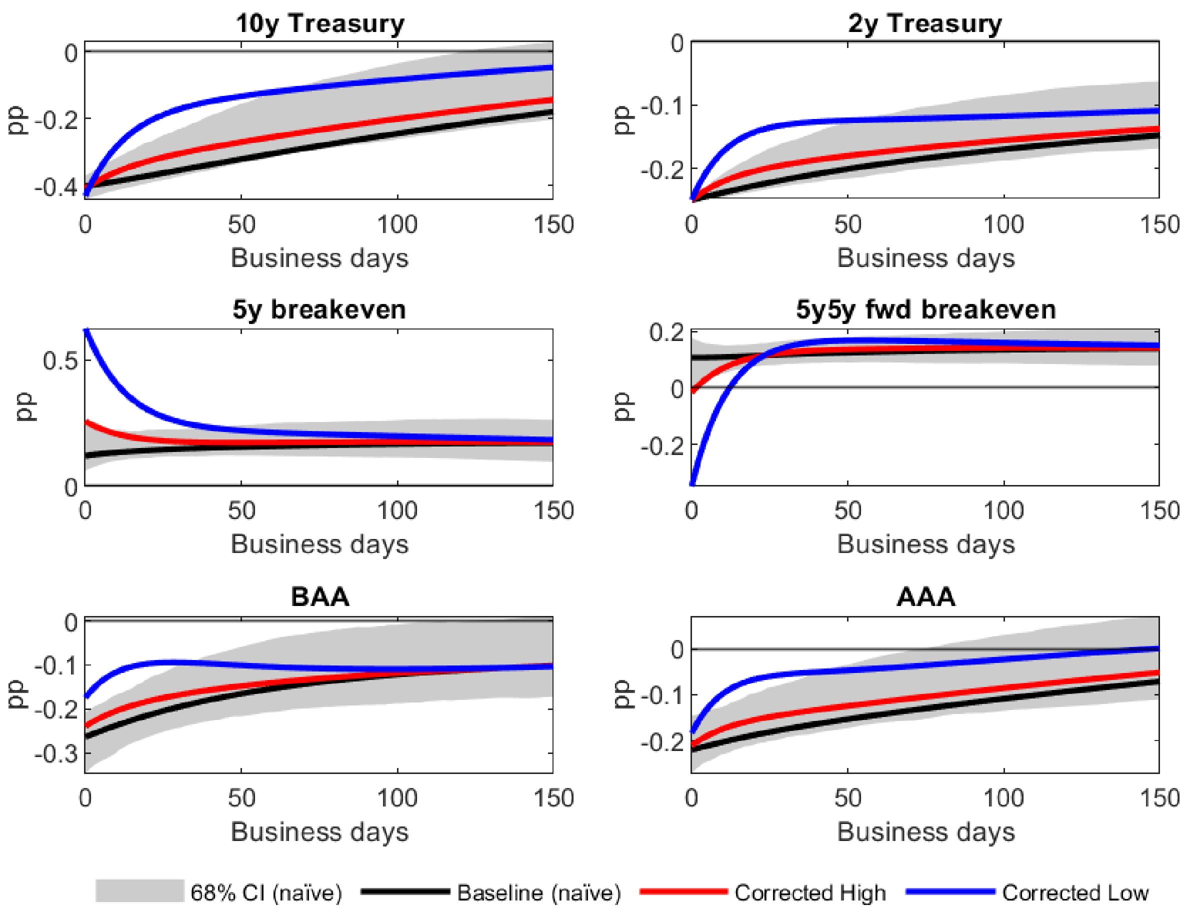
Finally, the linear mapping between Google Trends searches and attention is chosen for its simplicity and transparency. However, the core results below are robust to alternative mappings. Appendix E re-computes the same bias-corrected IRFs assuming several other mappings, and in all cases the bias correction substantially alters the results, with the same qualitative changes as those displayed below. Note that in all such mappings, I only make use of the *ratio* between (in)attention across regimes, so scaling by some baseline measure of attention leaves the results unchanged.

Results. Figure 1 shows the results. The uncorrected estimates following the specification in Wright (2012) are shown in black.[25] I normalize the shock such that it delivers a 25 basis point decline in the 2-year Treasury yield on impact. The corrected estimates for the high and low variance regimes are shown in red and blue respectively. Since Assumption 4 implies that the 2-year Treasury

[25] Note that while the VAR specification and identification scheme are identical to those in Wright (2012), the implementation differs. Wright (2012) uses a minimum-distance joint-fit estimator, while I use the spectral procedure discussed in Proposition 2, as that is the object whose properties are characterized formally above. The two estimators are equivalent in population when Assumption 1 holds and $\Delta\Omega_{H,L}$ is rank one, but may differ in finite samples.

yield is the same on impact in both regimes, I use the same normalization of the shock size in these corrected IRFs as in the uncorrected estimates.

Figure 1: Impulse response functions with and without the correction for endogenous attention.



Note: Impulse responses to an unconventional monetary policy shock identified via heteroskedasticity, following Wright (2012). The black line shows the uncorrected estimate assuming regime-invariant transmission (Assumption 1), with the shaded region indicating 68% bootstrap confidence intervals, constructed using the Kilian (1998)-adjusted bootstrap. The Kilian bias adjustment is also applied to the point estimates. The red line shows the bias-corrected impulse response for the high-attention regime (FOMC announcement days), and the blue line shows the bias-corrected response for the low-attention regime (non-announcement days). Bias correction follows the method in Section 4.1, imposing that the impact effect on the 2-year Treasury yield is attention-invariant (Assumption 4) and that attention amplifies the impact effect on BAA corporate bond yields (Assumption 5). The responses are normalized such that the 2-year Treasury yield falls by 25 basis points on impact in all cases. Sample: November 2008 to December 2015, excluding December 1 2008 (see text). The VAR is estimated with one lag on daily data.

For all variables, the uncorrected IRFs do not typically lie within the corrected IRFs for the two regimes. This is a consequence of the fact that the uncorrected IRFs are non-convex combinations of the true IRFs (Proposition 3). The bias-corrected IRFs are also not, in general, scaled versions of the uncorrected IRFs or of each other. Even though inattention only affects the impact matrix B in equation (1), it affects the impact of the shock on each of the variables in the VAR differentially. In the first period after the shock, the differential impacts then imply differential changes in the lags of the outcome vector, thus leading to sometimes very different dynamics across regimes.

The deviations from the uncorrected IRFs are also much smaller in the high-variance regime. For some questions, the high-variance/high-attention regime IRF may indeed be the relevant object: if the researcher is interested in the effect of policy decisions made on scheduled days then the high-attention regime is the appropriate one to consider. However, for other questions the researcher may be interested in monetary policy in general, such as from speeches that are not highly anticipated before they occur. In this case, the low-attention regime is also relevant. Indeed, this constitutes the vast majority of days in the sample. And in this case the uncorrected IRFs are notably biased for all variables. Moreover, even if transmission on scheduled announcement days is the relevant object for a researcher, I now go on to demonstrate that comparing corrected IRFs in high- and low-variance periods can shed light on the transmission channels behind the estimated effects, which is of interest even to those only considering shocks that arrive in a single regime.

Expectations and attention as drivers of policy transmission. In general, the large gaps between the corrected IRFs in the high and low variance regimes suggest that a substantial portion of the transmission of unconventional monetary policy comes through expectations, rather than mechanical balance-sheet or no-arbitrage effects. The horizon at which attention affects transmission, and whether it amplifies or dampens the shock's effects, vary across variables.

Treasury yields. On impact, unconventional monetary policy shocks have the same impact on the 2-year Treasury yield regardless of the regime. This holds by construction (Assumption 4). However, the impact is also the same across regimes for the 10-year Treasury yield, and this is not imposed by the bias-correction procedure. This is (i) reassuring that placing Assumption 4 on the impact effect on a Treasury yield is appropriate (see Robustness discussion below); and (ii) indicative that the impact of unconventional monetary policy on Treasury yields is largely independent of expectations, consistent with theories of preferred-habitat investors rebalancing portfolios (Vayanos and Vila, 2021).

However, after the initial day of the shock, the responses of Treasury yields begin to diverge between regimes. After 50 business days, the effects on the 2- and 10-year yields are approximately a third and a half smaller respectively in the low-variance regime. This suggests that without subsequent expectations updating, the initial portfolio-balance effects are largely transitory, echoing the (at the time controversial) conclusion of Wright (2012). Indeed, the bias correction reinforces Wright's conclusion: in the low-attention regime, which is the majority of the sample, the estimated effects decay even faster than in the uncorrected estimates. Attention causing greater persistence in Treasury yields is consistent, for example, with unconventional policy shocks in part signaling future policy actions by the Federal Reserve, as explored theoretically by Bhattarai et al. (2023).[26]

[26] Indeed, this is the logic used by Gürkaynak et al. (2005a) and the literature that follows them to separate monetary

Corporate bond yields. The AAA corporate bond yields follow similar patterns to Treasury yields: while there is a small amplifying effect of attention on impact, the IRFs for the two regimes diverge over the following days, with high attention amplifying the persistence of the response. This is as would be expected from portfolio rebalancing mechanisms, where high-quality corporate bond yields follow Treasury yields. The fact that attention amplifies the impact effect is also consistent with the arguments put forward in the discussion of Assumption 5 above, that when investors are more attentive to an expansionary monetary policy shock they revise their default probability expectations more strongly downwards, further reducing corporate bond yields.

For BAA corporate bond yields, attention substantially amplifies the effect of the shock on impact (>25% stronger impact effect in the high-variance regime, compared to 13% for AAA yields). This larger amplification from attention is again consistent with the default-expectations mechanism proposed above, as default probabilities are larger and more salient than for AAA-rated bonds. Similarly, it could also reflect a role for expectations in the risk-taking channel of monetary policy (Bauer et al., 2023): if investors paying attention to monetary policy increase their risk appetite, this would lower credit spreads through the excess bond premium (Gilchrist and Zakrajšek, 2012). In the low-attention regime, investors do not observe the change in monetary policy as precisely, and thus the amplification through risk appetite is weaker.

While this amplification effect initially grows, almost doubling the response one month after the shock, the impulse responses in the two regimes then converge. This suggests that the attention-sensitive component of this transmission is short-lived, whether it is through default expectations, risk premia, or both. At longer horizons, the effects on lower-rated investment-grade bond yields are largely driven by forces independent of monetary policy expectations. If default expectations are the key driver of the initial amplification, this could for example reflect investor learning about default risk at the firm level, which a range of information frictions would render sluggish (Mankiw and Reis, 2002; Angeletos et al., 2020).

Breakevens. The impulse responses for breakeven inflation rates show the most striking differences between high and low variance regimes. In the low-attention regime in particular, there is a pronounced rotation: the 5-year breakeven rises sharply while the 5-to-10-year forward breakeven falls.

While this might initially seem strange, it is actually evidence in favor of the intuitive argument given by Wright (2012) to explain the similar short-run rotation in breakevens in his results:[27]

policy shocks into "Target" and "Path/Forward Guidance" factors.

[27] See Haubrich et al. (2012) and D'Amico et al. (2018) for discussions of liquidity premia in inflation-linked Treasury markets. Abrahams et al. (2016) in particular find that variation in risk and liquidity premia were especially important for forward breakeven inflation during the financial crisis period.

There is a rotation of TIPS break-evens, with five-year break-evens rising and 5–10-year forward break-evens falling. A possible interpretation is that the stronger outlook for demand boosts the short-to-medium-run inflation outlook, but the fact that the LSAPs are overwhelmingly concentrated in nominal (rather than TIPS) securities has an offsetting effect, pushing longer term break-evens lower. Wright (2012), p.F460.

The key evidence in favor of this view is that when attention is high, the rotation in breakevens is a great deal smaller. In particular, the decline in forward breakevens is almost absent, consistent with expectations-updating in response to the expansionary shock pushing longer-term inflation expectations upward, partially offsetting the mechanical downward pressure posited by Wright (2012). The puzzling dynamics die away a month after the shock, again consistent with the driver being a short-run market friction, not a fundamental shift in expectations. At longer horizons breakeven inflation is estimated to increase after the shock.

The net effect on observed forward breakevens in the high-attention regime is small on impact. However, this small net effect reflects two large, opposing forces: a negative mechanical component (visible in isolation in the low-attention regime) and a positive expectations-updating component. The contribution of expectations-updating to policy transmission through inflation expectations is therefore substantially larger than one would infer from the observed breakeven movements alone.

Robustness. While the results presented here align with theoretical developments in the study of unconventional monetary policy, it should be reiterated that they rely on the Google Trends proxy for attention, and its mapping into the model object τ_s^*. Both of these steps are necessarily imperfect: despite its increasingly common use, Google Trends is only a noisy proxy for attention, and the mapping into the model is stylized. In Appendix E I therefore repeat the bias correction in Figure 1 for alternative calibrations of $r = \frac{1-\tau_L^*}{1-\tau_H^*}$, adjusting the Google Trends sample and the mapping into the model. All results are qualitatively consistent with Figure 1.

I also examine the sensitivity of the results to Assumption 4, which imposes that the impact effect of monetary policy on the 2-year Treasury yield is invariant to attention. The key threat to this assumption is that monetary policy shocks affect expectations of future policy or macroeconomic conditions on impact, and that this has an effect on Treasury yields beyond the mechanical portfolio rebalancing discussed above.

While ex ante this sounds plausible, two pieces of ex post evidence suggest that the concern is small. First, Figure 1 shows that the impact effect of monetary policy shocks on the 10-year Treasury yield is very similar between regimes, despite the attention invariance only being imposed on the 2-year yield. The corrected high-attention impact effect is within 5% of the low-attention impact, well within the bootstrap confidence intervals for the uncorrected estimates. This is consistent with portfolio-balance channels initially dominating in long-term assets, and the expectational effects previously mentioned being minimal. If expectational effects are this small for long-term Treasuries, they are even more likely to be small for shorter maturities.

Second, in Appendix E I relax Assumption 4, and instead choose the rotation ϕ^* to minimize the difference between the effects of attention on impact on the 2-year and 10-year Treasury yields. This approach is motivated by the hypothesis that Treasury markets of different maturities should be affected by similar forces. If expectational effects were large in these markets, this approach would reveal that, but this is not what I find. On impact, the response of the 2-year Treasury yield is less than 3% larger in the high- vs low-attention regime. Again, the results are consistent with Figure 1, suggesting that the main results derived under Assumption 4 are robust.

6 Conclusion

I have argued that endogenous information acquisition affects macroeconometric practice. Standard models of rational inattention, and a wealth of empirical evidence, predict that economic agents process more information about variables when their volatility rises. Since information processing affects the way those agents react to structural economic shocks, the transmission of those shocks depends on the variance of the process from which they were drawn. This challenges a key assumption required for structural vector autoregressions identified via heteroskedasticity, that the responses to structural shocks are constant, even as their distributions shift.

I show that ignoring this mechanism biases impulse responses from heteroskedasticity-based identification. However, I also show that the bias can be corrected: one method uses external attention proxies, another exploits variation across three or more variance regimes.

Beyond bias correction, these methods also yield new economic insights. Because attention reacts to the variance of the structural shocks, the true impulse responses to those shocks differ between variance regimes. The correction methods recover regime-specific impulse responses, and comparing them reveals how the updating of expectations contributes to the transmission of the structural shocks of interest.

I apply these methods to the transmission of unconventional monetary policy at the zero lower bound in the US, following and extending Wright (2012). I find that the corrected impulse responses show less persistence in the transmission to Treasury and corporate bond yields, and very different short-run responses of breakeven inflation rates, than those obtained from a standard identification that does not correct for endogenous information. Comparing responses across regimes reveals a powerful role for signaling channels of unconventional monetary policy, especially for long-term inflation expectations.

These insights are important for understanding the zero lower bound period, but also carry relevance for policymakers today. Understanding the transmission channels of unconventional monetary policy, and how much of that transmission depends on expectations, will be important as central banks embark on balance-sheet normalizations in the years to come. The most surprising implication is that if policymakers wish to conduct a balance-sheet normalization without strong effects on Treasury yields or corporate bond yields, they should not communicate in advance, so as to keep attention low. This would, however, come with greater short-run volatility in inflation breakevens. There may also be other reasons why communication would be desirable (Haldane et al., 2021): the results here simply suggest that this communication may also have costs in government and corporate bond markets.

Finally, the bias mechanism I discuss applies in principle to any heteroskedasticity-based identification, including Markov-switching and smooth-transition specifications. Developing practical correction methods for these more complicated settings is a promising avenue for future research. More broadly, the finding that a common econometric technique embeds implicit assumptions about information processing suggests that rational inattention may have unexplored implications for other areas of macroeconometrics.

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A Further Theoretical Discussion

A.1 A Microfoundation for Assumption 2

The expectations-augmented transmission equation (9) is intended to be general, encompassing a range of models in which shock realizations and perceptions may simultaneously affect transmission. In this appendix I give one possible microfoundation, which clarifies the distinction between the "mechanical" channel $\boldsymbol{\alpha}_1$ and the belief-dependent channel $\boldsymbol{\beta}_1$.

The economy is a limited-arbitrage asset market in the spirit of Vayanos and Vila (2021). The shock of interest is an unconventional monetary policy (asset-purchase) shock. It affects outcomes through two distinct routes: it changes the *net supply* of assets that private investors must absorb (a market-clearing effect that operates whatever investors believe), and it changes the *perceived value* of those assets to investors, e.g. if the shock is expected to affect macroeconomic conditions, which in turn feed back into asset prices.

Investors. There is a continuum of competitive investors $i \in [0,1]$ with CARA preferences and absolute risk aversion $\rho > 0$. Each allocates wealth between a riskless asset with gross return R_f and n risky long-term bonds with price vector q_t and payoff vector π_{t+1}. Holding the vector $x_{i,t} \in \mathbb{R}^n$ of risky assets, terminal wealth is

$$W_{i,t+1} \;=\; R_f W_{i,t} \;+\; x_{i,t}'\big(\pi_{t+1} - R_f q_t\big). \tag{41}$$

Payoffs are

$$\pi_{t+1} \;=\; \bar{\pi} \;+\; \delta\, \varepsilon_{1,t} \;+\; u_{t+1}, \qquad u_{t+1} \sim \mathcal{N}(0, \Sigma_u), \tag{42}$$

where $\delta \in \mathbb{R}^n$ loads each asset's payoff on the policy shock, and u_{t+1} is residual payoff risk, independent of $\varepsilon_{1,t}$.

Information. Investors have the information structure described in Section 3: each investor does not observe $\varepsilon_{1,t}$, but does observe a private signal $z_{i,t}$ as in equation (13). They also know the variance regime s (Assumption 3). Finally, I make the simplifying assumption here that investors form their beliefs about $\varepsilon_{1,t}$ before observing asset prices.

By Lemma 1, the posterior belief about the policy shock has mean $\hat{\varepsilon}_{i,t} = E_i[\varepsilon_{1,t} \mid z_{i,t}, s] = \tau_s^* z_{i,t}$, and has variance equal to $\theta/2$ whenever attention is positive.[28] The perceived payoff distribution is

[28] This is a standard feature of Gaussian rational-inattention problems: agents process more information when σ_s^2 is larger, exactly enough to hold posterior uncertainty fixed. I assume attention is interior throughout this appendix.

therefore $\pi_{t+1} \mid z_{i,t}, s \sim \mathcal{N}\big(\bar{\pi} + \delta\hat{\varepsilon}_{i,t},\, \Sigma_\pi\big)$, with the regime-invariant perceived covariance

$$\Sigma_\pi \;=\; \Sigma_u \;+\; \delta\delta'\,\tfrac{\theta}{2}. \tag{43}$$

Optimal demand. With CARA preferences and Gaussian payoffs, investor i chooses $x_{i,t}$ to maximize the certainty-equivalent $x'_{i,t}(\mu_{i,t} - R_f q_t) - \frac{\rho}{2} x'_{i,t}\Sigma_\pi x_{i,t}$, where $\mu_{i,t} = \bar{\pi} + \delta\hat{\varepsilon}_{i,t}$. The first-order condition gives the standard mean-variance demand

$$x_{i,t} \;=\; \frac{1}{\rho}\,\Sigma_\pi^{-1}\big(\bar{\pi} + \delta\hat{\varepsilon}_{i,t} - R_f q_t\big). \tag{44}$$

Because Σ_π is common across investors and regimes by (43), aggregating (44) over i and applying the law of large numbers ($\int_0^1 \hat{\varepsilon}_{i,t}\, di = \bar{E}_t[\varepsilon_{1,t}] = \tau_s^* \varepsilon_{1,t}$, as in (20)) yields aggregate demand

$$\bar{x}_t \;=\; \frac{1}{\rho}\,\Sigma_\pi^{-1}\big(\bar{\pi} + \delta\,\bar{E}_t[\varepsilon_{1,t}] - R_f q_t\big). \tag{45}$$

Supply and market clearing. Asset supply is given by the fixed quantity $\bar{S}$, minus any central bank purchases. The net supply that investors must hold is therefore

$$S_t \;=\; \bar{S} \;-\; \phi\,\varepsilon_{1,t}, \qquad \phi \in \mathbb{R}^n_{\geq 0}, \tag{46}$$

where ϕ_j is the incidence of the (expansionary) purchase shock on market j: a positive $\varepsilon_{1,t}$ shrinks the available asset supply, with the reduction distributed across assets according to ϕ. Market clearing $\bar{x}_t = S_t$, using (45) and (46), gives

$$R_f q_t \;=\; \bar{\pi} \;+\; \delta\,\bar{E}_t[\varepsilon_{1,t}] \;-\; \rho\,\Sigma_\pi\big(\bar{S} - \phi\varepsilon_{1,t}\big), \tag{47}$$

so that the equilibrium price vector is

$$q_t \;=\; \underbrace{\tfrac{1}{R_f}\big(\bar{\pi} - \rho\Sigma_\pi\bar{S}\big)}_{\text{constant}} \;+\; \tfrac{\rho}{R_f}\,\Sigma_\pi\phi\,\varepsilon_{1,t} \;+\; \tfrac{1}{R_f}\,\delta\;\bar{E}_t[\varepsilon_{1,t}]. \tag{48}$$

Mapping to equation (9). If prices q_t are the observables,[29] equation (48) is exactly the contemporaneous part of the transmission equation (9), with

$$\boldsymbol{\alpha}_1 \;=\; \frac{\rho}{R_f}\,\Sigma_\pi\phi, \qquad\qquad \boldsymbol{\beta}_1 \;=\; \frac{1}{R_f}\,\delta. \tag{49}$$

[29] Note that in the application in Section 5.3 the observables are yields, not prices, so the signs of each coefficient would be reversed relative to what is presented here.

Using $\bar{E}_t[\varepsilon_{1,t}] = \tau_s^* \varepsilon_{1,t}$, the impact vector of the shock in regime s is $\boldsymbol{\alpha}_1 + \tau_s^* \boldsymbol{\beta}_1$, reproducing the regime-dependent impact matrix (21). The remaining structural shocks enter through the constant term and the covariance Σ_u, and the lag polynomial $A(L)$ arises from the dynamics of the state variables ($\bar{S}$, $\bar{\pi}$, R_f).

Discussion. The loading $\boldsymbol{\alpha}_1$ is the price-pressure response required to clear the market when the asset supply changes. It operates through risk-bearing (ρ, Σ_π) and supply incidence (ϕ) alone, and contains no expectation of the shock. It is constant across regimes because optimal attention holds the perceived covariance Σ_π at the regime-invariant level (43): that is, the regime-invariance of $\boldsymbol{\alpha}_1$ is a result here rather than an assumption. All regime dependence in the impact vector comes through $\tau_s^* \boldsymbol{\beta}_1$, as posited in (9).

Since the components of equation (9) are derived rather than assumed here, we can additionally study how they change with the deep parameters of the model. In particular, as $\rho \to 0$ (risk-neutral arbitrage, perfectly elastic demand) or $\phi \to 0$ (purchases do not alter the net asset supply), $\boldsymbol{\alpha}_1 \to 0$ and (48) collapses to $q_t \approx \frac{1}{R_f} \delta \, \bar{E}_t[\varepsilon_{1,t}]$, in which prices respond only to the perceived shock. In this context, the presence of the true shock $\varepsilon_{1,t}$ alongside the perceived shock in (9) is therefore a consequence of limited arbitrage: $\boldsymbol{\alpha}_1$ is the price-pressure term that vanishes in frictionless settings but is generically present once risk-bearing capacity is finite.

We can also similarly consider structural interpretations of Assumption 4. The attention-invariant restriction $\beta_{1,k} = 0$ holds if and only if $\delta_k = 0$: i.e. if asset k's payoff does not load on the policy shock. Such an asset still responds to the shock mechanically, through ϕ_k and the cross-market covariance in $\Sigma_\pi \phi$, but not through revisions in beliefs about fundamentals. In Section 5.3 I impose this on the 2-year Treasury yield: the structural interpretation of this assumption is therefore that the response of this yield to asset purchases is driven by supply and price pressure, with negligible re-pricing of medium-run macroeconomic fundamentals at that maturity.

This microfoundation also helps to give a general intuition for what "mechanical" channels are in (9). The loading $\boldsymbol{\alpha}_1$ collects all effects that operate through equilibrium or accounting constraints rather than through inference about the shock. Limited-arbitrage price pressure is the leading example here, but the same logic covers the repricing of floating-rate or inflation-indexed claims, cash-flow effects of rate changes operating through budget constraints, and collateral revaluations that tighten or relax position constraints.

Alternatively, one could consider frictionless markets with a fixed share λ of fully informed and $1 - \lambda$ of inattentive investors. This would deliver the same reduced form, with $\boldsymbol{\alpha}_1$ proportional to λ and $\boldsymbol{\beta}_1$ to $1 - \lambda$.

Generality of equation (9). The asset-market model above is one concrete microfoundation, but the structure of (9) is considerably more general. Whether a shock affects outcomes independently of agents' perceptions of it (i.e. whether $\boldsymbol{\alpha}_1$ and $\boldsymbol{\beta}_1$ are both non-zero) depends on the structure of the model: specifically, on whether the shock enters resource constraints, budget constraints, or market-clearing conditions, in addition to the first-order conditions of inattentive agents. Effects operating through the former are what I term "mechanical" ($\boldsymbol{\alpha}_1$): they are pinned down by the realized shock whatever agents believe. Effects operating through the latter are belief-dependent ($\boldsymbol{\beta}_1$): they enter only through agents' perceptions, and so are scaled by attention τ_s^*.

Equation (9) thus arises whenever a shock has both kinds of effect, which is the case across a wide range of standard frameworks. Several examples illustrate the point.

- *Real business cycle models.* A productivity shock enters the aggregate resource constraint directly, affecting allocation possibilities whatever agents believe (mechanical, $\boldsymbol{\alpha}_1$). It also enters the consumption-saving and labor-supply first-order conditions through household perceptions of the shock (belief-dependent, $\boldsymbol{\beta}_1$). The output response depends on both.
- *Open-economy models.* A terms-of-trade shock enters the trade balance and import bill directly through the budget and market-clearing constraints (mechanical, $\boldsymbol{\alpha}_1$). It also enters expenditure-switching and pricing decisions through expectations (belief-dependent, $\boldsymbol{\beta}_1$). The net-export response depends on both.
- *New Keynesian models.* In medium-scale models such as Smets and Wouters (2007), a fraction of firms and labor unions cannot optimally reset prices/wages each period, but instead follow a mechanical indexing rule. An inflationary shock affects the indexed prices and wages of non-adjusting firms and unions regardless of any belief about the shock (mechanical, $\boldsymbol{\alpha}_1$). It also affects the reset prices and wages of adjusting firms and unions, which are forward-looking and depend on expectations of current and future conditions (belief-dependent, $\boldsymbol{\beta}_1$). The aggregate price- and wage-inflation responses depend on both.
- *Heterogeneous information.* Equation (9) also arises in economies populated by a fixed share λ of fully informed agents and a share $1-\lambda$ of inattentive agents. The informed respond to the realized shock and the inattentive to their perceptions of it, delivering (9) with $\boldsymbol{\alpha}_1 \propto \lambda$ and $\boldsymbol{\beta}_1 \propto (1-\lambda)$. Heterogeneity of this kind is employed in e.g. Macaulay (2021) and Broer et al. (2026), and evidence has been provided in, among others, Link et al. (2023) and Boccanfuso and Neri (2025). Note that here the "mechanical" loading $\boldsymbol{\alpha}_1$ runs through the (correct) beliefs of the informed rather than through a constraint; the reduced form is nonetheless identical to (9).

A.2 Comparison to Identification via Non-Gaussianity

To further highlight which features of the rational inattention model are critical for the identification failure characterized in Proposition 1, it is instructive to compare to identification strategies that use non-Gaussianity of structural shocks (e.g. Lanne and Lütkepohl, 2010; Jarociński, 2024). Such identification schemes have an equivalent of Assumption 1, that the impact matrix B must be constant across the whole support of the shock distribution. Intuitively, the transmission of a shock must be proportional to its size, whether the shock comes from the tails or the center of the shock distribution.

Importantly, this assumption is a statement about varying *realizations* of the structural shock, while Assumption 1 concerns changes in the shock *distribution*. In standard rational inattention models, as in the model set out above, attention choices are made using knowledge of the shock's distribution, but before the shock is realized. As a result, there can be no variation of attention between shocks from the center or tail of a fat-tailed distribution, and the identification failure derived here cannot apply to identification via non-Gaussianity.[30] Even though non-Gaussianity and heteroskedasticity-based identification fall into the same family of statistical identification methods (Lewis, 2025), they therefore differ in their susceptibility to endogenous attention, because the attention mechanism exclusively relies on properties of the distribution of shocks, not their realizations. The robustness of identification via non-Gaussianity can be seen as a benefit in many situations, though it also means that such strategies cannot be used to elicit the effects of variable attention on shock transmission, as is done in Sections 4 and 5 of the paper.

While the basic identification failure studied in this paper does not therefore operate in SVARs identified via non-Gaussianity, there are two additional channels through which rational inattention may affect such identification schemes, which future research could potentially explore.

First, as discussed extensively by Jarociński (2024), identification via non-Gaussianity often relies heavily on the mutual independence of structural shocks, not just the invariance of the impact vector to shock realizations. In many cases, optimal information processing over multiple variables can lead to shock perceptions which are correlated with each other, even if the shocks themselves are independent (Kőszegi and Matějka, 2020). This occurs, for example, if the agent only wishes to forecast some combination of variables, rather than each variable independently: a simple example is that households may want to forecast the real wage more than they want to know nominal wages and inflation separately from one another (Kamdar and Ray, 2024). That means the most efficient way to process information is to obtain a signal that combines several variables, rather than independent signals for each component variable. As a result, if (as posited in Assumption 2) shock transmission runs partly through shock perceptions, rational inattention may undermine the

[30]Note that this could cease to be true if other models of information choice were adopted, in which attention can vary with shock realizations.

independence assumption even if the structural shocks themselves are fully independent. Note that this only applies if attention affects more than one shock, beyond the single-shock case studied here.

Second, identification via heteroskedasticity or non-Gaussianity is typically performed in the context of linear SVARs. In the model outlined in this paper, rational inattention does not disrupt that linearity, as the optimal signal structure is such that it preserves the linear mapping from shocks to actions. However, this linear policy function is sensitive to departures from Gaussianity in the shock processes. If shocks have fat tails, optimal signal structures and the resulting policy functions are typically highly nonlinear (Matějka, 2016; Ellison and Macaulay, 2021), in ways that make log-linearizations poor approximations to aggregate dynamics. Rational inattention in non-Gaussian contexts may therefore introduce a specific form of misspecification into linear SVARs.

A.3 Comparison to High-Frequency Event Studies

If there is no variation in attention (i.e. Assumption 1 is satisfied), then $\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1 = \boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1$. By equation (23), $\Delta\boldsymbol{\Omega} = (\sigma_H^2 - \sigma_L^2)(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1)(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1)'$ and is rank 1, and so the leading eigenvector is proportional to $\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1$. If, in addition, all shocks except the policy shock of interest are turned off, as in a narrow announcement window, then $\boldsymbol{\Omega}_H = \sigma_H^2(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1)(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1)'$ is also rank 1. A high-frequency event study regression uses $\boldsymbol{\Omega}_H$ directly to extract $\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1$. This is the sense in which "*Identification through heteroskedasticity collapses to the event-study methodology in the limiting case that the announcement windows contain only the shocks that we wish to identify — that is, when the variances of all other shocks are negligible.*" (Wright, 2012, p. F457).

However, this equivalence breaks down in the presence of rational inattention, which as shown in Section 3 breaks Assumption 1. In that case, $\Delta\boldsymbol{\Omega} = \sigma_H^2 (\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1)(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1)' - \sigma_L^2 (\boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1)(\boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1)'$ is rank 2, and the leading eigenvector is proportional to $(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1) + k \cdot (\boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1)$ (Proposition 2). That is, the estimated $\hat{b}_1$ from the SVAR identified via heteroskedasticity is a (often non-convex) combination of regime-specific impact vectors. Proposition 2 is unaffected by the variance of other (non-policy) shocks, because they have equal variance across regimes and so cancel out in the difference $\Delta\boldsymbol{\Omega}$. The event study regression, in contrast, only makes use of the 'events' (i.e. high-variance periods), and thus still obtains $\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1$ if non-policy shocks are 0. Even in this limit, the two identification methods therefore fail to coincide.

Although the simple equivalence between methods stated above by Wright no longer holds, there is a situation in which the methods still coincide: if we add to the "no non-policy shocks" condition a second limit, that policy shocks have 0 variance in the low-attention regime (i.e. if $\sigma_L^2 \to 0$). In this case, $\Delta\boldsymbol{\Omega} \to \sigma_H^2(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1)(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1)'$, and all identification in the SVAR comes from the high-attention regime. Formally, in equation (52) below the constant term becomes 0, and so the root $k \to 0$, so we obtain $\hat{b}_1 \propto \boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1$ as in the event study. To use the monetary policy shock example, to match event-study regressions we would have to obtain a set of periods in which it is believed that there were no monetary policy shocks at all: a substantially harder requirement than "no non-policy shocks," which is typically justified by the narrow intraday window used in event studies.

This highlights that event-study regressions, common in the identification of monetary policy shocks, do not suffer from an inattention-driven bias. However, they are only able to recover the shock transmission in the high-attention regime. In some contexts, this may be the object of interest. Section 5.3, however, discusses why it may often be advantageous to also study shocks in the low-attention regime. More broadly, this analysis highlights that estimates from event studies may not easily generalize to shocks that arrive when attention is low.

B Proofs

B.1 Proof of Lemma 1

The first-order condition for the problem (15) is:

$$-\sigma_s^2 + \frac{\theta}{2} \cdot \frac{1}{1-\tau_s^*} = 0. \tag{50}$$

Rearranging gives the first case in equation (16). Positivity requires $\sigma_s^2 > \theta/2$. When this condition holds (i.e. at the interior solution for $\tau_{i,t}$) the derivative of τ_s^* with respect to σ_s^2 is

$$\frac{d\tau_s^*}{d\sigma_s^2} = \frac{\theta}{2\sigma_s^4} > 0. \tag{51}$$

B.2 Proof of Proposition 2

Let

$$v_H \equiv \boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1, \qquad v_L \equiv \boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1,$$

and define

$$\Delta\boldsymbol{\Omega} = \sigma_H^2 v_H v_H' - \sigma_L^2 v_L v_L'.$$

Step 1: Reduction to the span of $\{v_H, v_L\}$.

For any vector x such that $x'v_H = x'v_L = 0$, we have

$$\Delta\boldsymbol{\Omega} x = \sigma_H^2 v_H (v_H' x) - \sigma_L^2 v_L (v_L' x) = 0.$$

Hence, any eigenvector of $\Delta\boldsymbol{\Omega}$ associated with a nonzero eigenvalue must lie in the two-dimensional subspace $span\{v_H, v_L\}$. It follows that any such eigenvector can be written as

$$x = v_H + kv_L$$

for some scalar $k \in \mathbb{R}$.

Step 2: Characterization of eigenvectors.

Let $x = v_H + kv_L$. Then

$$\Delta\boldsymbol{\Omega}x = \sigma_H^2 v_H(v_H' x) - \sigma_L^2 v_L(v_L' x) = \sigma_H^2 v_H(C_H + kD) - \sigma_L^2 v_L(D + kC_L),$$

where

$$C_s \equiv v_s' v_s, \qquad D \equiv v_H' v_L.$$

If x is an eigenvector with eigenvalue μ, then $\Delta\boldsymbol{\Omega}x = \mu x$, which implies

$$\sigma_H^2(C_H + kD) = \mu,$$
$$-\sigma_L^2(D + kC_L) = \mu k.$$

Eliminating μ yields

$$-\sigma_L^2(D + kC_L) = k\sigma_H^2(C_H + kD),$$

which rearranges to

$$(\sigma_H^2 D)\,k^2 + (\sigma_H^2 C_H + \sigma_L^2 C_L)\,k + (\sigma_L^2 D) = 0. \tag{52}$$

If $D \neq 0$, this quadratic has two real roots, each corresponding to an eigenvector direction of $\Delta\boldsymbol{\Omega}$ in $span\{v_H, v_L\}$.

Step 3: Identification of the "naïve" estimator.

The standard "naïve" spectral procedure sets $\hat{b}_1$ proportional to the eigenvector associated with the largest (positive) eigenvalue of $\Delta\boldsymbol{\Omega}$. Hence,

$$\hat{b}_1 \propto v_H + kv_L,$$

where k is the root of the above quadratic corresponding to the largest eigenvalue.

Step 4: Generic distinctness from regime-specific impacts.

If v_H and v_L are collinear, then $\Delta\boldsymbol{\Omega}$ is rank one and all eigenvectors associated with nonzero eigenvalues are proportional to both v_H and v_L. If $v_H \perp v_L$, then $D = 0$ and the unique eigenvector associated with the positive eigenvalue is proportional to v_H.

Outside of these knife-edge cases, the eigenvector $v_H + kv_L$ is not proportional to either v_H or v_L. Therefore, the uncorrected estimator $\hat{b}_1$ generically differs from both regime-specific impact vectors, implying that the resulting impulse responses correspond to a pseudo-parameter rather than the true structural impact in either regime.

B.3 Proof of Corollary 1

By regime invariance of $A(L)$, the MA matrices $\mathbf{\Psi}_h$ do not depend on s. For each regime s, the IRF is $IRF_s(h) = \mathbf{\Psi}_h v_s$. The uncorrected estimator satisfies $\hat{b}_1 \propto v_H + kv_L$, hence

$$\widehat{IRF}(h) \propto \mathbf{\Psi}_h \hat{b}_1 = \mathbf{\Psi}_h(v_H + kv_L) = IRF_H(h) + kIRF_L(h),$$

which is the first claim. The normalized expression follows by dividing both sides by the impact normalization $e'_m(IRF_H(0) + kIRF_L(0))$. The final statement follows because (outside the knife-edge cases) the vectors $IRF_H(h)$ and $IRF_L(h)$ are not collinear for generic h, so their linear combination is not proportional to either one.

B.4 Proof of Proposition 3

Since v_H and v_L are non-zero outside the degenerate case $\beta_1 = 0$, we have $C_H, C_L > 0$. The proposition concerns the roots of the quadratic (52). I take $D \neq 0$ throughout, so that (52) is a genuine quadratic; the orthogonal case $D = 0$ is treated separately at the end. Let a, b, and c be the coefficients on k^2, k, and the constant in (52) respectively.

Step 1: Prove the roots are real. The discriminant of (52) is

$$\begin{aligned} b^2 - 4ac &= (\sigma_H^2 C_H + \sigma_L^2 C_L)^2 - 4\,\sigma_H^2\sigma_L^2 D^2 \\ &= (\sigma_H^2 C_H - \sigma_L^2 C_L)^2 + 4\,\sigma_H^2\sigma_L^2\big(C_H C_L - D^2\big). \end{aligned} \tag{53}$$

The first term in (53) is a square and hence non-negative. For the second, the Cauchy–Schwarz inequality applied to v_H and v_L gives $D^2 = (v'_H v_L)^2 \leq (v'_H v_H)(v'_L v_L) = C_H C_L$, so $C_H C_L - D^2 \geq 0$. Both terms are therefore non-negative, the discriminant is non-negative, and the two roots of (52) are real.

Step 2: Prove the roots share a common sign. Denote the two roots of (52) as k_1, k_2. Applying Vieta's formulas, their product is

$$k_1 k_2 = \frac{c}{a} = \frac{\sigma_L^2 D}{\sigma_H^2 D} = \frac{\sigma_L^2}{\sigma_H^2} > 0, \tag{54}$$

implying that the two real roots are non-zero and have the same sign.

Step 3: The common sign. Again by Vieta, the sum of the roots is

$$k_1 + k_2 = -\frac{b}{a} = -\frac{\sigma_H^2 C_H + \sigma_L^2 C_L}{\sigma_H^2 D}. \tag{55}$$

The numerator $\sigma_H^2 C_H + \sigma_L^2 C_L$ is strictly positive, since $\sigma_H^2, \sigma_L^2 > 0$ and $C_H, C_L > 0$. Hence the sign of the sum is $-\operatorname{sign}(\sigma_H^2 D) = -\operatorname{sign}(D)$. Two real numbers of common sign (Step 2) whose sum has sign $-\operatorname{sign}(D)$ must each individually have sign $-\operatorname{sign}(D)$. Therefore

$$\operatorname{sign}(k_1) = \operatorname{sign}(k_2) = -\operatorname{sign}(D) = -\operatorname{sign}\big(v_H' v_L\big). \tag{56}$$

Since both roots share the same sign, the root corresponding to the largest eigenvalue of $\Delta\boldsymbol{\Omega}$ (i.e. k) satisfies equation (28).

The orthogonal case. If $D = 0$ then $a = c = 0$, so (52) degenerates to $b\,k = 0$ with $b > 0$, giving the single root $k = 0$. The naïve estimate is then $\hat{b}_1 \propto v_H$: with $v_H \perp v_L$, the matrix $\Delta\boldsymbol{\Omega} = \sigma_H^2 v_H v_H' - \sigma_L^2 v_L v_L'$ has v_H and v_L as its two eigenvectors with eigenvalues $\sigma_H^2 C_H > 0$ and $-\sigma_L^2 C_L < 0$ respectively, so the largest eigenvalue selects v_H exactly. This is the boundary between the convex and non-convex regions, consistent with (56) as $D \to 0$.

C Relationship with Specification Tests

Assumption 1, which I have shown above is affected by rational inattention, is often simply imposed in practical applications. However, it is possible to test it in some contexts. I therefore discuss if and how the bias due to rational inattention could be detected in these tests.

Multiple regimes and overidentifying restrictions. When more than two variance regimes are available, Assumption 1 implies that all pairwise differences $\boldsymbol{\Omega}^{(j)} - \boldsymbol{\Omega}^{(k)}$ must be rank one and share a common eigenvector. These restrictions are overidentifying: they imply that the eigenvectors of $\boldsymbol{\Omega}^{(j)} - \boldsymbol{\Omega}^{(k)}$ coincide across all j, k.

Under endogenous attention, these restrictions are generically violated. The reduced-form covariance differences are given by

$$\boldsymbol{\Omega}^{(j)} - \boldsymbol{\Omega}^{(k)} = \sigma_j^2(\boldsymbol{\alpha}_1 + \tau_j^*\boldsymbol{\beta}_1)(\boldsymbol{\alpha}_1 + \tau_j^*\boldsymbol{\beta}_1)' - \sigma_k^2(\boldsymbol{\alpha}_1 + \tau_k^*\boldsymbol{\beta}_1)(\boldsymbol{\alpha}_1 + \tau_k^*\boldsymbol{\beta}_1)', \tag{57}$$

which does not share a common rank-one factorization across j, k unless the vectors $\{\boldsymbol{\alpha}_1 + \tau_s^*\boldsymbol{\beta}_1\}_{s=1}^S$ are all collinear. Thus, even with arbitrarily many regimes, there is in general no regime-invariant impact vector that rationalizes the joint behavior of the reduced-form covariance matrices.

In principle, the overidentifying restrictions that arise when more than two variance regimes are present could be used to test for whether the VAR parameters are constant across regimes. Such tests could detect the specific failure of Assumption 1 generated by endogenous information acquisition (see e.g. Rigobon, 2003, for an example of such a parameter-stability test). However, in practice, these tests often lack power in the sample sizes typical of macroeconomic applications. As Lewis (2021, 2022) highlights, identification via heteroskedasticity is frequently 'weak,' meaning that any statistical signal from variance changes is often drowned out by sampling noise. Economically significant shifts in the impact matrix $\mathbf{B}$ due to endogenous attention may still therefore escape rejection in parameter-stability tests.

Rank-one restrictions. If the impact matrix $\mathbf{B}$ is invariant across regimes and only the variance of a single structural shock changes, the difference in reduced-form covariance matrices $\Delta\boldsymbol{\Omega}_{j,k} \equiv \boldsymbol{\Omega}^{(j)} - \boldsymbol{\Omega}^{(k)}$ is rank one. Equivalently, $\Delta\boldsymbol{\Omega}_{j,k}$ has exactly one nonzero eigenvalue in population.

In contrast, under the data-generating process described above, $\Delta\boldsymbol{\Omega}_{j,k}$ is generically of rank *two* whenever $\boldsymbol{\alpha}_1 + \tau_j^*\boldsymbol{\beta}_1$ and $\boldsymbol{\alpha}_1 + \tau_k^*\boldsymbol{\beta}_1$ are linearly independent. Hence, the rank-one condition required for heteroskedasticity-based identification fails in population, as a direct consequence of endogenous information acquisition.

While formal tests for the rank of the covariance difference matrix exist (e.g. Robin and Smith, 2000; Chen and Fang, 2019), they face similar limitations to the multiple-regime based tests described above. The rational inattention mechanism explored in this section implies a rank-two structure for $\Delta\boldsymbol{\Omega}_{j,k}$, but the second non-zero eigenvalue may still be small relative to the estimation error of the covariance matrices. In this context, a failure to statistically reject the rank-one null in Robin-Smith style tests does not confirm that the impact vector is constant. It may simply reflect that the attention-driven distortion, while sufficient to bias the point estimate, is difficult to distinguish from sampling noise using second moments alone. In the application to monetary policy in Section 5, I do indeed find that $\Delta\boldsymbol{\Omega}_{j,k}$ has an eigenvalue structure consistent with the rational inattention model, but that these eigenvalues are generally small and estimated somewhat imprecisely.

D Bias Correction Simulations

Correction method 1. Figure 2 displays the results from a VAR(1) in two variables, with two variance regimes and true parameters given by:

$$\mathbf{A}_1 = \begin{pmatrix} 0.8, 0.1 \\ 0.25, 0.7 \end{pmatrix}, \ \alpha_1 = \begin{pmatrix} 1 \\ 0.5 \end{pmatrix}, \ \beta_1 = \begin{pmatrix} -0.7 \\ 0 \end{pmatrix}, \ \mathbf{b}_2 = \begin{pmatrix} 0.2 \\ 0.7 \end{pmatrix}, \ (\sigma_H^2, \sigma_L^2) = (4, 1), \ \theta = 0.75. \tag{58}$$

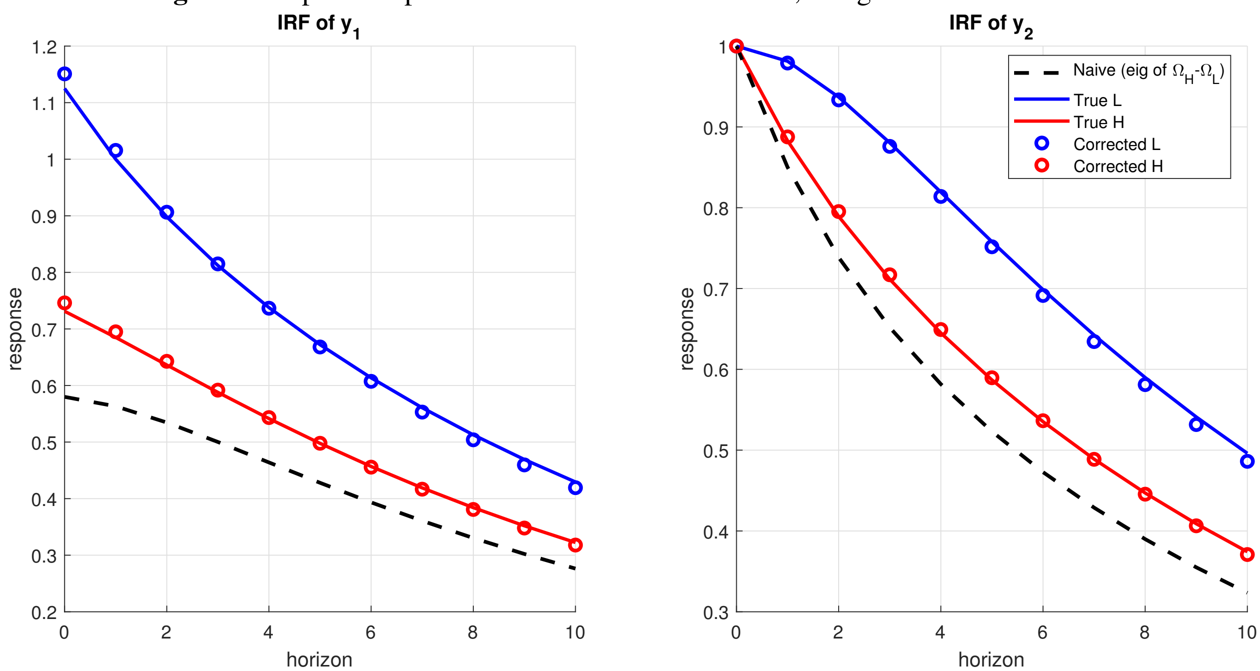
The solid red and blue lines show the true impulse responses to the shock of interest in the high- and low-variance regimes respectively. The dashed black lines show the impulse responses estimated on a 2,500-observation simulated sample if the researcher does not correct for the bias derived in Proposition 2, and if the high-variance regime occurs with probability 0.15. The red and blue circles show the impulse responses estimated on the same simulated data, using the bias correction method in Section 4.1, with Assumptions 4 and 5 chosen correctly, and r measured exactly. Since y_2 is the variable invariant to attention on impact, I normalize the shock so that the impact is 1 on this variable in all cases. The correction is able to very closely match the true underlying responses in each regime, while the uncorrected estimates are far from either true response. The length of the simulation sample is chosen to match that in the application of Section 5.3.

Figure 2 shows the results in the ideal situation, in which Assumptions 4 and 5 hold exactly, and the attention proxy (and so the calibration of r) is perfect. Figures 3 and 4 explore the sensitivity of the method to deviations in the accuracy of Assumption 4 and the calibration of r.

First, consider the sensitivity to Assumption 4. I maintain the parameters listed in (58), except I vary the second element of β_1 between -0.35 and 0.35. At each of 21 equally spaced points in this range, I draw 1,000 random samples of 2,500 observations each. I then compute standard and corrected IRFs, with the correction assuming that the second element of β_1 is 0, as in Figure 2. I find the maximum absolute error between each IRF and the true IRF at which they are aiming. Figure 3a then plots the median of these maximum-absolute-errors across simulations. The median maximum-absolute-error for the corrected IRFs is minimized when $\beta_1(2) = 0$, but it is still substantially below the equivalent errors from the uncorrected estimation for a wide range around that. Indeed, for small deviations from the strict attention-invariance in Assumption 4, the errors in the corrected IRFs do not increase substantially: if the impact effect of attention on y_2 is 30% as strong as on y_1, then the median maximum-absolute-error increases by 8% (high-attention regime) and 14% (low-attention regime) respectively.

Another way to show this is to plot the improvement in estimates resulting from the correction. Figure 4a plots the results from the same simulations as Figure 3a, but instead of the median maximum-absolute-error, it shows the percentage improvement in that error over the uncorrected

Figure 2: Impulse response functions in simulated data, using correction method 1.

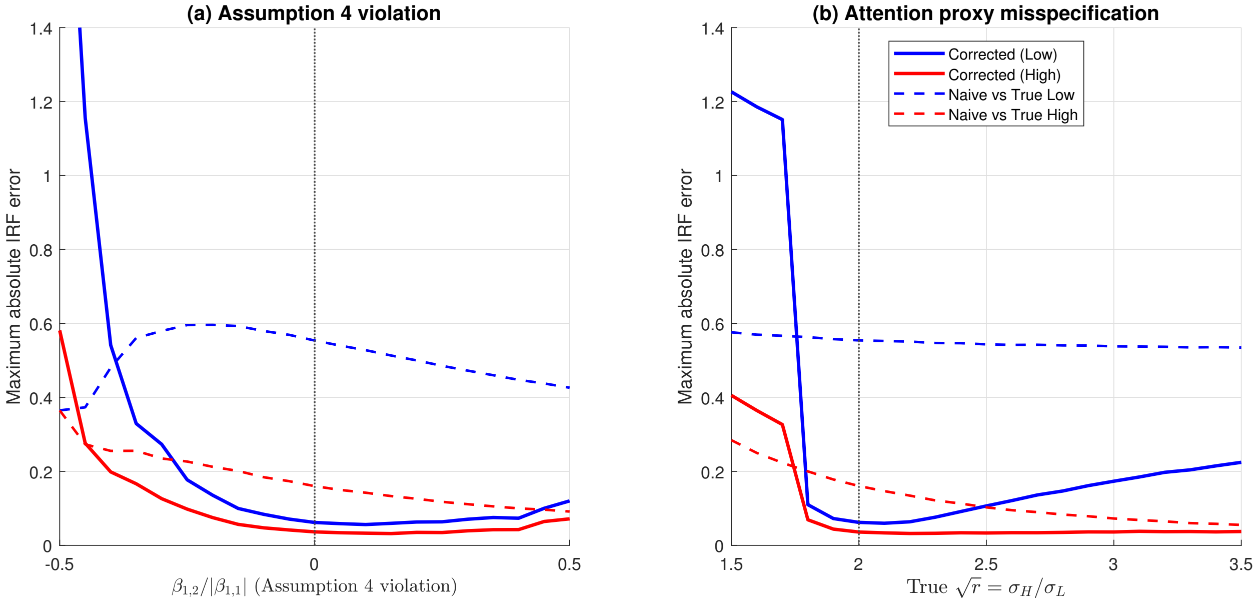


Note: Impulse responses to structural shock 1 in simulated data from a VAR(1) with two variables and two variance regimes. True parameters are given in Appendix D, equation (58). The solid red and blue lines show the true impulse responses in the high-variance and low-variance regimes, respectively. The dashed black line shows the estimate obtained by applying standard heteroskedasticity-based identification to a simulated sample of 2,500 observations, with the high-variance regime occurring with probability 0.15. Red and blue circles show bias-corrected estimates using the method in Section 4.1, with correct specification of Assumptions 4 and 5 and exact measurement of the inattention ratio r. Responses are normalized such that the impact on y_2 equals one.

case. When Assumption 4 is exactly true, the correction removes more than three quarters of the error from the uncorrected estimates (more than 88% in the low-variance regime where uncorrected errors are larger). However, even when y_2 is not exactly attention-invariant, the improvements remain substantial.

Figures 3b and 4b repeat the exercise, but for robustness to mismeasurement of r. In this case the true $\sqrt{r}$ is varied between 1.5 and 3.5. Again, at each grid point 1,000 simulated samples are drawn, and in each sample the standard and corrected IRFs are computed, with the correction (wrongly) assuming $\sqrt{r} = 2$ as in (58). Figure 3b plots the median maximum-absolute-error across simulations, and Figure 4b plots the percentage improvement in that error over the uncorrected estimates. As with Assumption 4, small deviations from the assumptions used in the correction make the errors larger, but not substantially, until the true r goes a long way from the measured value. This is especially an issue if the true r gets too small, as then the dynamics become quite similar in the two regimes.

Figure 3: Sensitivity of correction method 1 to Assumption 4 and r measurement: median maximum-absolute-error across simulations.



Note: Sensitivity of correction method 1 to violations of Assumption 4 (panel a) and misspecification of the inattention ratio r (panel b). In panel (a), simulated samples are drawn from the system with parameters described in (58), except for $\beta_1(2)$, which is varied between -0.35 and 0.35. In panel (b), $\beta_1(2)$ is fixed at 0, but the true $\sqrt{r}$ is varied between 1.5 and 3.5. At each set of parameters, 1,000 random samples are drawn, and both standard and corrected impulse responses are computed. The maximum absolute error between each impulse response and the corresponding true impulse response is computed for each sample over horizons 0-10. The plots show the medians of these maximum-absolute-errors across the simulations. Solid lines show errors for the bias-corrected estimates; dashed lines show errors for uncorrected estimates that ignore endogenous attention.

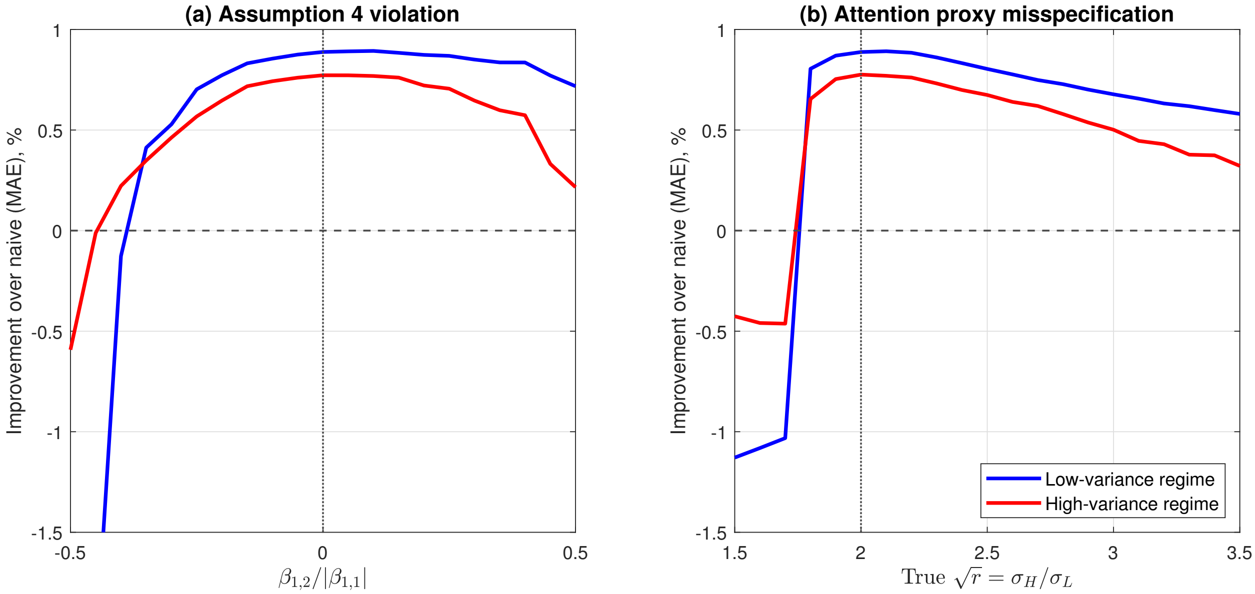
Correction method 2. Figure 5 displays the results from a VAR(1) in three variables, with three variance regimes and true parameters given by:

$$\mathbf{A}_1 = \begin{pmatrix} 0.6, 0.3, 0.05 \\ 0.05, 0.5, 0.1 \\ 0, 0.15, 0.55 \end{pmatrix}, \ \alpha_1 = \begin{pmatrix} 0.4 \\ -0.1 \\ 0.2 \end{pmatrix}, \ \beta_1 = \begin{pmatrix} 0.6 \\ 0.5 \\ -0.3 \end{pmatrix}, \ (\mathbf{b}_2, \mathbf{b}_3) = \begin{pmatrix} 0.2, -0.1 \\ 0.3, 0.15 \\ -0.1, 0.25 \end{pmatrix},$$
$$(\sigma_H^2, \sigma_M^2, \sigma_L^2) = (8, 1.2, 0.6), \ \theta = 1, \ (p_H, p_M, p_L) = (0.7, 0.2, 0.1). \tag{59}$$

where p_s is the probability of regime s occurring. As with the simulation above, the regimes are drawn i.i.d., and the variances of the other structural shocks not being studied are normalized to 1. The IRFs plotted below normalize the structural shock so that it delivers a unit increase in the first variable in the VAR in all cases.

The solid colored lines show the true impulse responses to the shock of interest in each regime. The dashed black lines show the impulse responses estimated on a 2,500-observation simulated sample if the researcher does not correct for the bias derived in Proposition 2. The colored circles show the impulse responses estimated using the bias correction method in Section 4.2. While the results are not as close to the true impulse responses as those from method 1 (Figure 2), especially

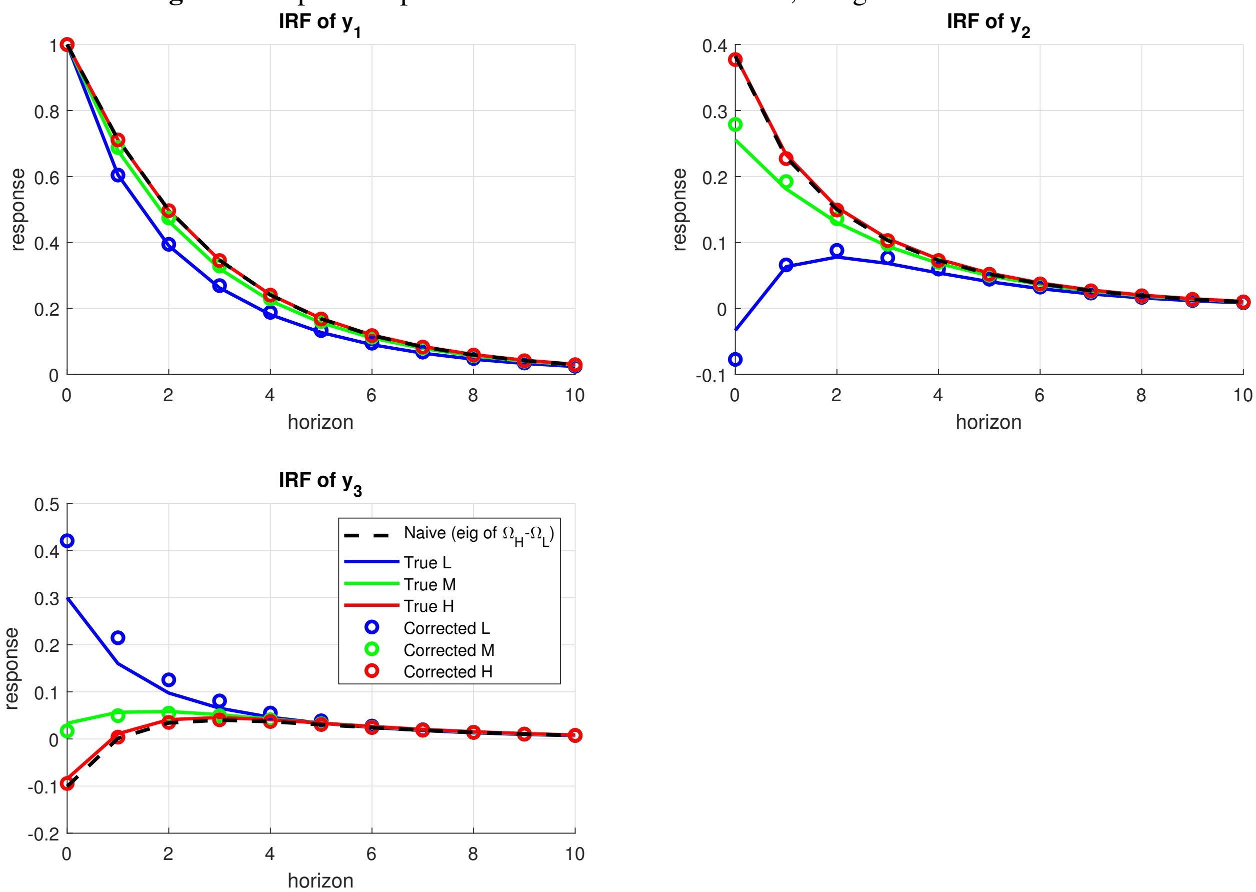
Figure 4: Sensitivity of correction method 1 to Assumption 4 and r measurement: percentage improvement in median maximum-absolute-error across simulations.



Note: Sensitivity of correction method 1 to violations of Assumption 4 (panel a) and misspecification of the inattention ratio r (panel b). In each panel, simulations are drawn for varying parameters as in Figure 3 (see associated note for details). The median maximum-absolute-error across simulations is computed for standard and bias-corrected impulse responses as in Figure 3. The lines plotted here show the percentage improvement in this median maximum-absolute-error in the bias-corrected impulse responses, relative to the uncorrected impulse responses. Specifically, they show $(\text{medmax}(\text{uncorrected}) - \text{medmax}(\text{corrected}))/\text{medmax}(\text{uncorrected})$. Positive numbers reflect a reduction in median maximum-absolute-errors due to the bias correction, negative numbers reflect an increase.

for the lowest attention regime, this method requires fewer assumptions on the role of attention in each variable's transmission, and still delivers impulse responses close to the true responses in each regime.

Figure 5: Impulse response functions in simulated data, using correction method 2.



Note: Impulse responses to structural shock 1 in simulated data from a VAR(1) with three variables and three variance regimes. True parameters are given in equation (59). Solid red, green, and blue lines show the true impulse responses in the high-, medium-, and low-variance regimes, respectively. The dashed black line shows the estimate obtained by applying standard heteroskedasticity-based identification to a simulated sample of 2,500 observations, with regime probabilities $(p_H, p_M, p_L) = (0.7, 0.2, 0.1)$. Colored circles show bias-corrected estimates using the method in Section 4.2. Responses are normalized such that the impact on y_1 equals one.

E Section 5 Robustness

Table 4 repeats Table 2, using an alternative Google Trends index reflecting searches for the term "monetary policy." Although quantitative magnitudes vary, the key result that search volume is higher on announcement days is robust.

Figures 6 and 7 show the bias-corrected IRFs, as in Figure 1, for alternative mappings between Google Trends data and τ_L^*, τ_H^*. In Figure 6, I maintain the linear mapping used for Figure 1, but calculate average Google Trends volume in each regime after trimming the sample to exclude the largest and smallest 1% of observations, which yields $r = 2.04$. In Figure 7, I assume a linear mapping between $\tau_{i,t}$ and *log* Google Trends volume, which yields $r = \frac{\log(100)-\log(50.1)}{\log(100)-\log(86.3)} = 4.69$. In both cases, the core qualitative results of Figure 1 remain: correcting for the rational-inattention bias substantially alters the estimated effects of monetary policy shocks. In particular, the low-variance

Table 4: Attention to monetary policy: FOMC vs. Non-FOMC Days

Sample	Observations		Mean Search Index		Difference	t-statistic
	FOMC	Non-FOMC	FOMC	Non-FOMC		
Wright (2008–2011)	23	1,039	53.461	38.223	15.238	3.056
ZLB (2008–2015)	56	2,537	50.342	37.251	13.090	4.290
Full (2006–2019)	113	4,816	49.525	35.421	14.104	6.444

Notes: Google Trends search index for "Monetary Policy" constructed using the Eichenauer et al. (2022) methodology. FOMC days are scheduled Federal Open Market Committee announcement days. The t-statistic tests the null hypothesis that mean attention is equal across FOMC and non-FOMC days.

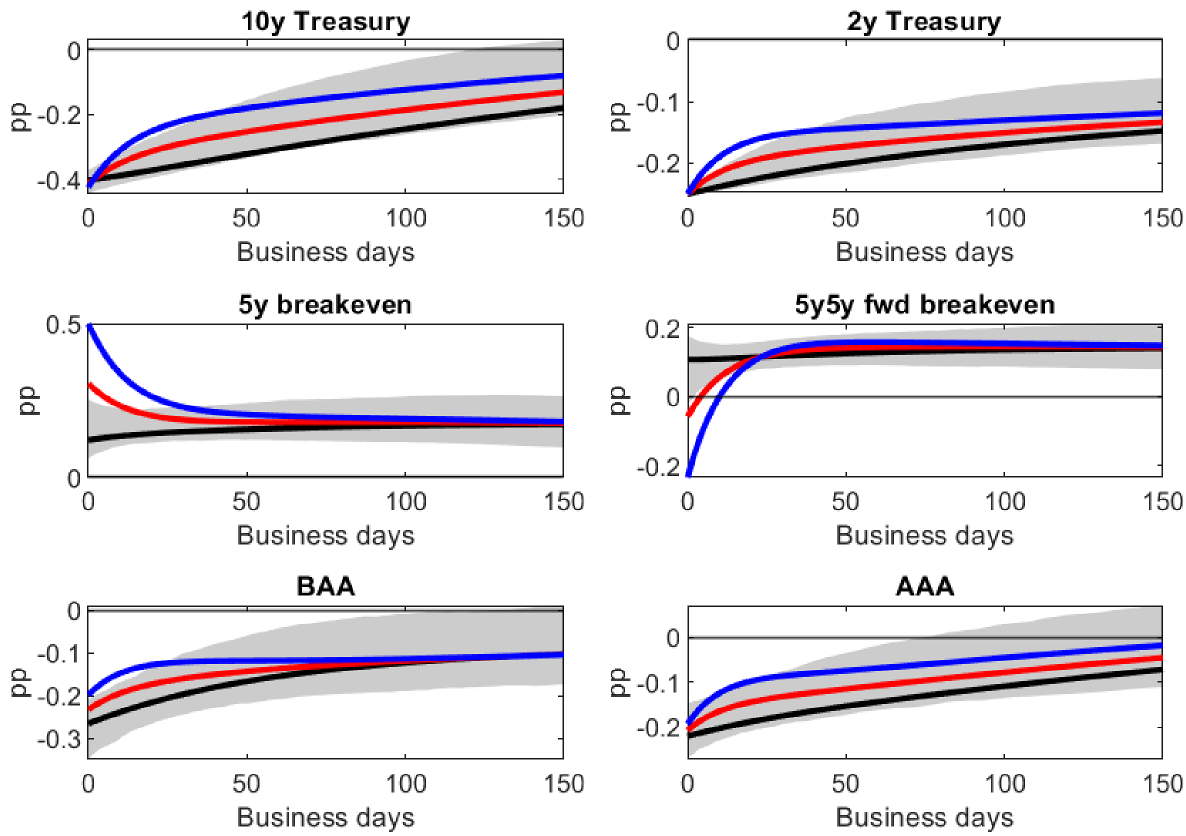
regime features substantially less persistent transmission to Treasury yields, and an initial sharp rotation in breakevens. Finally, Figure 8 shows the bias-corrected IRFs from an alternative procedure in which ϕ^* is chosen to minimize the gap between impact effects of attention on the two Treasury yields, as an alternative to the strict invariance of the 2-year yield in Assumption 4.

Specifically, let $v_H(\phi) = w_H(\phi)/\sigma_H$ and $v_L(\phi) = w_L(\phi)/\sigma_L$ denote the unscaled regime-dependent impact vectors implied by a candidate rotation ϕ, via equations (32)-(33) and the variance ratio r. I select the rotation ϕ^* that minimizes the transmission gap across regimes for both the 2-year and 10-year Treasury yields. That is, I replace equation (36) with:

$$\phi^* = \arg\min_{\phi} \sum_{k \in \{2y, 10y\}} \left(\frac{v_{H,k}(\phi)}{v_{L,k}(\phi)} - 1 \right)^2 \tag{60}$$

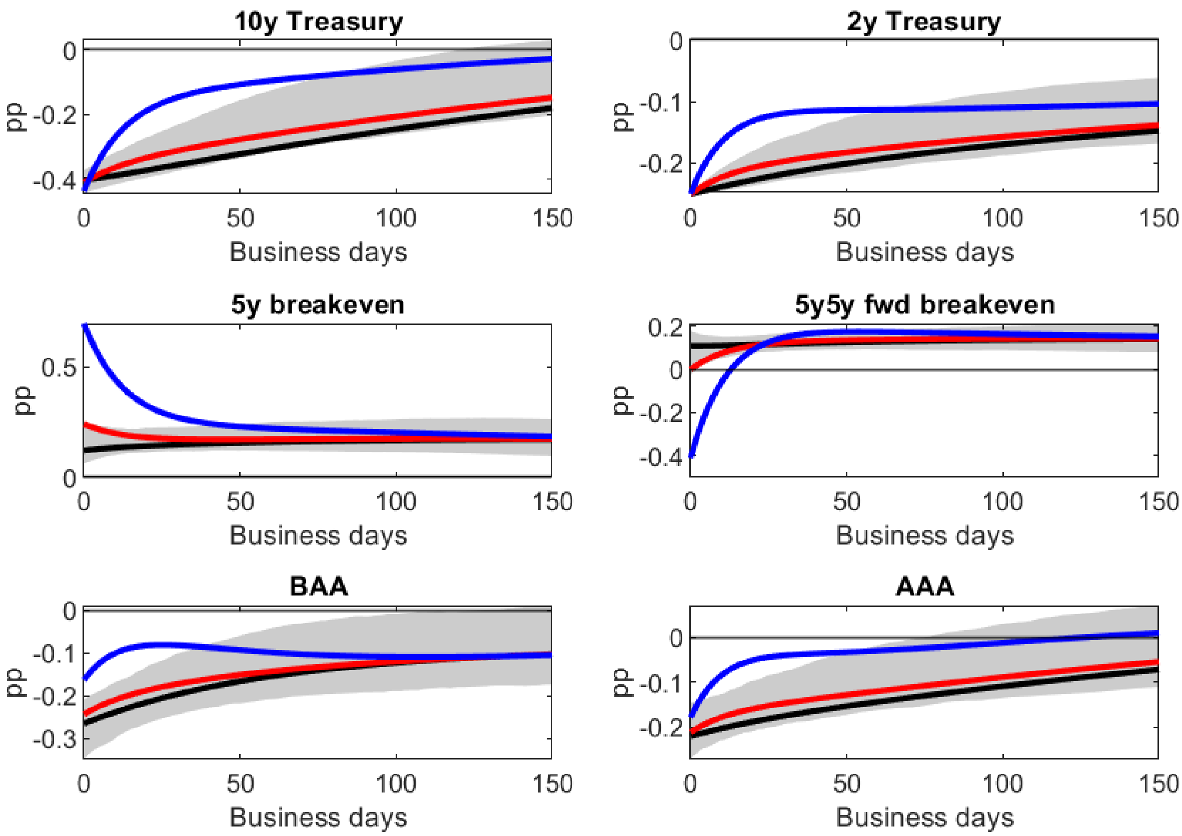
Using squared fractional differences ensures that the identification is not disproportionately driven by the maturity with the largest absolute response. This therefore seeks the structural model most consistent with the hypothesis that the Treasury curve is driven on impact by equally sized attention effects across maturities, without forcing any single maturity to be perfectly invariant. To make the extent of departure from Assumption 4 clear, I normalize the corrected IRFs for both regimes by the same scaling factor, chosen such that the impact is a 25bp fall in the 2-year Treasury yield on average across the regimes, rather than normalizing each impulse response individually. The results are qualitatively identical to Figure 1, and also quantitatively extremely similar.

Figure 6: Impulse response functions with and without the correction for endogenous attention, with trimmed Google Trends sample as the attention measure.



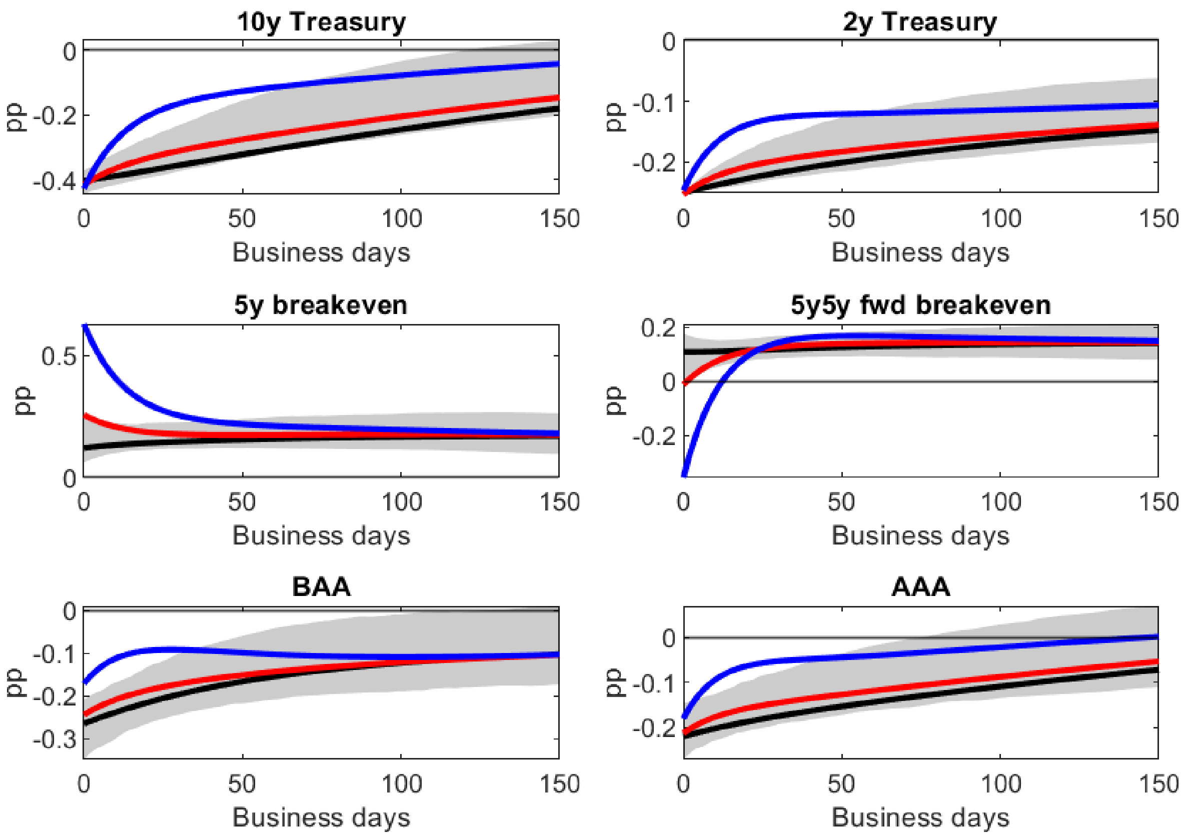
Note: Impulse responses to an unconventional monetary policy shock identified via heteroskedasticity, following Wright (2012). The black line shows the estimate assuming regime-invariant transmission (Assumption 1), with the shaded region indicating 68% bootstrap confidence intervals, constructed using the Kilian (1998)-adjusted bootstrap. The Kilian bias adjustment is also applied to the point estimates. The red line shows the bias-corrected impulse response for the high-attention regime (FOMC announcement days), and the blue line shows the bias-corrected response for the low-attention regime (non-announcement days). Bias correction follows the method in Section 4.1, imposing that the impact effect on the 2-year Treasury yield is attention-invariant (Assumption 4) and that attention amplifies the impact effect on BAA corporate bond yields (Assumption 5). The inattention ratio r is measured using the Google Trends data described in Section 5.2, after trimming the top and bottom 1% of the observations. The responses are normalized such that the 2-year Treasury yield falls by 25 basis points on impact in all cases. Sample: November 2008 to December 2015, excluding December 1 2008 (see text). The VAR is estimated with one lag on daily data.

Figure 7: Impulse response functions with and without the correction for endogenous attention, with log Google Trends sample as the attention measure.



Note: Impulse responses to an unconventional monetary policy shock identified via heteroskedasticity, following Wright (2012). The black line shows the estimate assuming regime-invariant transmission (Assumption 1), with the shaded region indicating 68% bootstrap confidence intervals, constructed using the Kilian (1998)-adjusted bootstrap. The Kilian bias adjustment is also applied to the point estimates. The red line shows the bias-corrected impulse response for the high-attention regime (FOMC announcement days), and the blue line shows the bias-corrected response for the low-attention regime (non-announcement days). Bias correction follows the method in Section 4.1, imposing that the impact effect on the 2-year Treasury yield is attention-invariant (Assumption 4) and that attention amplifies the impact effect on BAA corporate bond yields (Assumption 5). The inattention ratio r is measured using the log of the Google Trends series described in Section 5.2. The responses are normalized such that the 2-year Treasury yield falls by 25 basis points on impact in all cases. Sample: November 2008 to December 2015, excluding December 1 2008 (see text). The VAR is estimated with one lag on daily data.

Figure 8: Impulse response functions with and without the correction for endogenous attention, with minimum distance criterion for selecting ϕ^*.



Note: Impulse responses to an unconventional monetary policy shock identified via heteroskedasticity, following Wright (2012). The black line shows the estimate assuming regime-invariant transmission (Assumption 1), with the shaded region indicating 68% bootstrap confidence intervals, constructed using the Kilian (1998)-adjusted bootstrap. The Kilian bias adjustment is also applied to the point estimates. The red line shows the bias-corrected impulse response for the high-attention regime (FOMC announcement days), and the blue line shows the bias-corrected response for the low-attention regime (non-announcement days). Bias correction follows the method for Figure 1, except that Assumption 4 is replaced with equation (60). The responses are normalized such that the 2-year Treasury yield falls by 25 basis points on impact in the uncorrected case, and on average across the two corrected cases. Sample: November 2008 to December 2015, excluding December 1 2008 (see text). The VAR is estimated with one lag on daily data.

F Correction Method 2 Application

This appendix provides a practical application of correction method 2, as set out in Section 4.2. Specifically, I estimate the effects of labor-market shocks on the stock market and monetary policy. To do this, I begin with a simple VAR over four daily variables: the *3-month T-bill rate* and the *10-year Treasury yield* summarize monetary policy at short and long maturities; both are expressed as daily basis-point changes. The *Dow Jones Industrial Average (DJIA)* summarizes aggregate stock market performance, and is expressed as a daily % change. Finally, the *Dow Jones U.S. Business Training & Employment Agencies (DJUSBE)* sub-index, also expressed as a daily % change, provides a close link to labor-market fluctuations specifically. The sample runs from May 17 2010 (when DJUSBE historical data becomes available) to December 31 2019 (before any COVID-induced market disruptions).[31] Since all variables are expressed as daily changes, their autocorrelation is low,[32] and I proceed with a VAR(0).

To identify the effects of a structural labor-market shock, I construct three regimes.

1. High variance of labor market shocks: BLS Employment Situation Report (i.e. "the jobs report") release days.
2. Medium variance of labor market shocks: ADP National Employment Report release days. Before 2022, this was a noisy signal of the payroll statistics in the BLS report, released several days before the BLS report.
3. Low variance of labor market shocks: days with no BLS Employment Situation Report or ADP National Employment Report release, or any other major labor market data release. 'Other major releases' are defined as any day containing a release of (i) DOL unemployment insurance weekly claims numbers; (ii) BLS Job Openings and Labor Turnover Survey (JOLTS) data; (iii) BLS Employment Cost Index (ECI) data; (iv) FOMC announcement.[33]

Do the regimes feature different shock variances? Formal tests of whether these regimes feature sufficiently different shock variances for identification, such as those developed in Lütkepohl et al. (2021), are typically available only for two variance regimes, not three. The tests in Lütkepohl et al. (2021) also over-reject the null of no identification when the number of variables becomes large: the authors find that "*the magnitude of the distortions is quite substantial*" (p.13) for VARs with 5 variables, so the issue is likely to also be present with the four-variable system considered here. I

[31] I do not continue the sample after these disruptions because of a change in the ADP report methodology in 2022, which made it less comparable to the BLS jobs report, and thus less useful for my identification strategy.

[32] 1st-order autocorrelations are -0.076 (3m T-bill, 1st diff.), -0.028 (10y Treasury, 1st diff.), -0.038 (DJIA, % change), and -0.024 (DJUSBE, % change).

[33] While these are not strictly labor-market releases, I drop them because the variances of the T-bill and Treasury rates are both substantially higher on FOMC days, affecting the assumption made throughout the paper that only one shock systematically changes variance between regimes.

can, however, at least consider some simple diagnostics as reassurance.

First, note that for each regime s, the trace of the reduced-form variance-covariance matrix $\mathbf{\Omega}^{(s)}$ gives the total innovation variance across all shocks. In this sample, these traces are 28.0 (low regime), 33.4 (medium regime), and 59.2 (high regime). Overall, then, shock volatility is increasing from low to medium to high regimes, consistent with the identifying assumptions on the volatility of labor market shocks.

However, the trace of $\mathbf{\Omega}^{(s)}$ also contains the variance of non-labor market shocks. For a second check, note therefore that if we have a candidate direction d for the impact of labor market shocks, we can project the reduced-form variance-covariance matrices onto that direction using $d'\mathbf{\Omega}^{(s)}d/d'd$. Ex-ante, the only candidate d available is the uncorrected estimate denoted $\hat{b}_1$ in Proposition 2. Ex-post, we will also have corrected estimates for each of the three regimes. Though all are subject to uncertainty, and $\hat{b}_1$ may even be a non-convex combination of the true impact vectors (Proposition 3), all candidate directions yield labor-market shock variances that increase from the low to the medium to the high regime, again consistent with the identifying assumptions.[34]

Finally, consider the difference between regimes' reduced-form covariance matrices, $\Delta\mathbf{\Omega}_{j,k}$, where regime j is assumed in the identifying assumptions to have greater variance of labor-market shocks than regime k. If those identifying assumptions hold, such a difference matrix should feature a large positive eigenvalue, reflecting the increased variance of labor-market shocks in regime j relative to k. This pattern is present for the regime comparisons here: the largest-magnitude eigenvalues of $\Delta\mathbf{\Omega}_{M,L}$ and $\Delta\mathbf{\Omega}_{H,L}$ are large and positive (6.73 and 23.5 respectively).[35]

Implementation and results. I implement the estimation described in Section 4.2, with the DJUSBE percent change chosen as the anchor variable for normalization (that is, the impact of the shock on DJUSBE is normalized to equal 1 in the high-variance regime). This gives an estimated $\tau_M^* = 0.595$, and estimated impact vectors presented in Figure 9 panel a (plotted alongside the estimates from an uncorrected identification via heteroskedasticity for comparison). Panel b plots the same results, rescaled on a regime-by-regime basis so that the shock causes a unit increase in DJUSBE in all regimes. Both are informative.

First, note that the uncorrected estimates lie outside of the range of corrected estimates for the 3m T-bill and the 10y Treasury yield. As in the monetary application in Section 5.3, the uncorrected estimates are therefore a *non-convex* combination of regime-specific impacts. As confirmation

[34]With $d = \hat{b}_1$, the implied variances are 21.5 (L), 27.8 (M), and 45.1 (H). With $d = \hat{v}_L$ (where $\hat{v}_s$ is the corrected impact vector for regime s obtained from correction method 2) these are 5.01 (L), 5.71 (M), and 7.63 (H); with $d = \hat{v}_M$ they are 20.10, 26.45, and 42.50; with $d = \hat{v}_H$ they are 21.38, 27.76, and 44.88. Note that since $\hat{v}_s$ is estimated from the model assuming the identifying assumptions hold, these latter diagnostics should be treated as a consistency check not independent validation of the identification.

[35]As in Section 5.1, the difference matrices also contain a negative eigenvalue (with a smaller magnitude than the positive one: -1.20 and -1.45 for $\Delta\mathbf{\Omega}_{M,L}$ and $\Delta\mathbf{\Omega}_{H,L}$ respectively), as predicted when attention varies across regimes.

of this, the angle between the estimated $(\boldsymbol{\alpha}_1 + \tau_H^* \boldsymbol{\beta}_1)$ and $(\boldsymbol{\alpha}_1 + \tau_L^* \boldsymbol{\beta}_1)$ has cosine $0.425 > 0$, indicating non-convexity via the result in Proposition 3.

Next, we move on to the impacts themselves, starting with DJUSBE. The shock causes a substantially greater increase in employment services companies' stock prices when attention is high (panel a). This is intuitive: when investors pay more attention to labor market shocks, those shocks will have greater effects on the asset prices of the most affected firms.

That greater shock transmission also appears in the effects on the aggregate stock market (DJIA). Again, greater attention amplifies the effect of the shock, and indeed expectation effects are the large majority of the shock's transmission. Panel b reveals that this is true even when we rescale by regime: on low-attention days, a shock causing a 1% rise in the stock prices of employment services firms causes a 0.068% rise in the Dow Jones Industrial Average. On medium-attention days, a 1% DJUSBE shock causes a 0.167% rise in the DJIA, and on high-attention days this rises further to 0.182%. Attention is therefore important for the shock's overall effects, and for the links between the most labor-market exposed firms and the wider stock market.

The 10-year Treasury yield shows similar patterns to the DJIA, consistent with positive labor-market news leading investors to expect monetary tightening. Shocks that imply a 1% rise in DJUSBE (panel b) generate 2.06, 3.47, and 3.68bp increases in the 10-year yield in low, medium, and high variance regimes respectively. Again, increased attention to the labor market increases the transmission of labor-market shocks, in this case to policy. This is consistent, for example, with the results on monetary policy reactions to broader asset price movements in Rigobon and Sack (2003).

Finally, the 3-month T-bill rate is perhaps the most interesting result from this exercise. The uncorrected estimate implies that short rates had almost no reaction to labor-market shocks that move asset prices. With no further information, it would be tempting to ascribe this to the fact that approximately half of the sample is from the ZLB period. However, the 3-month T-bill rate had a mean absolute daily change of 0.64bp over 2010-2015, smaller than that in 2016-2019 (1.48bp), but not 0. And, indeed, the corrected estimates find that expectational effects are in fact strong and positive, just as they are with the 10-year Treasury yield. This suggests that when investors (and policymakers) are paying attention, labor market shocks increase expectations of labor market performance, and this increases the short rate relative to where it would have been without this change in expectations.

This positive expectational effect is unobserved in the uncorrected estimates (and in the high-variance regime which it most resembles) because it is offsetting a large negative mechanical effect, in the same way that expectational effects on breakeven inflation are masked in the monetary policy application in Section 5.3. This non-expectational effect could come from numerous sources. I offer three here, which all would predict the result in Figure 9, that the low-attention regime features a negative response of short rates, and close-to-zero response of long rates, to labor-market shocks.

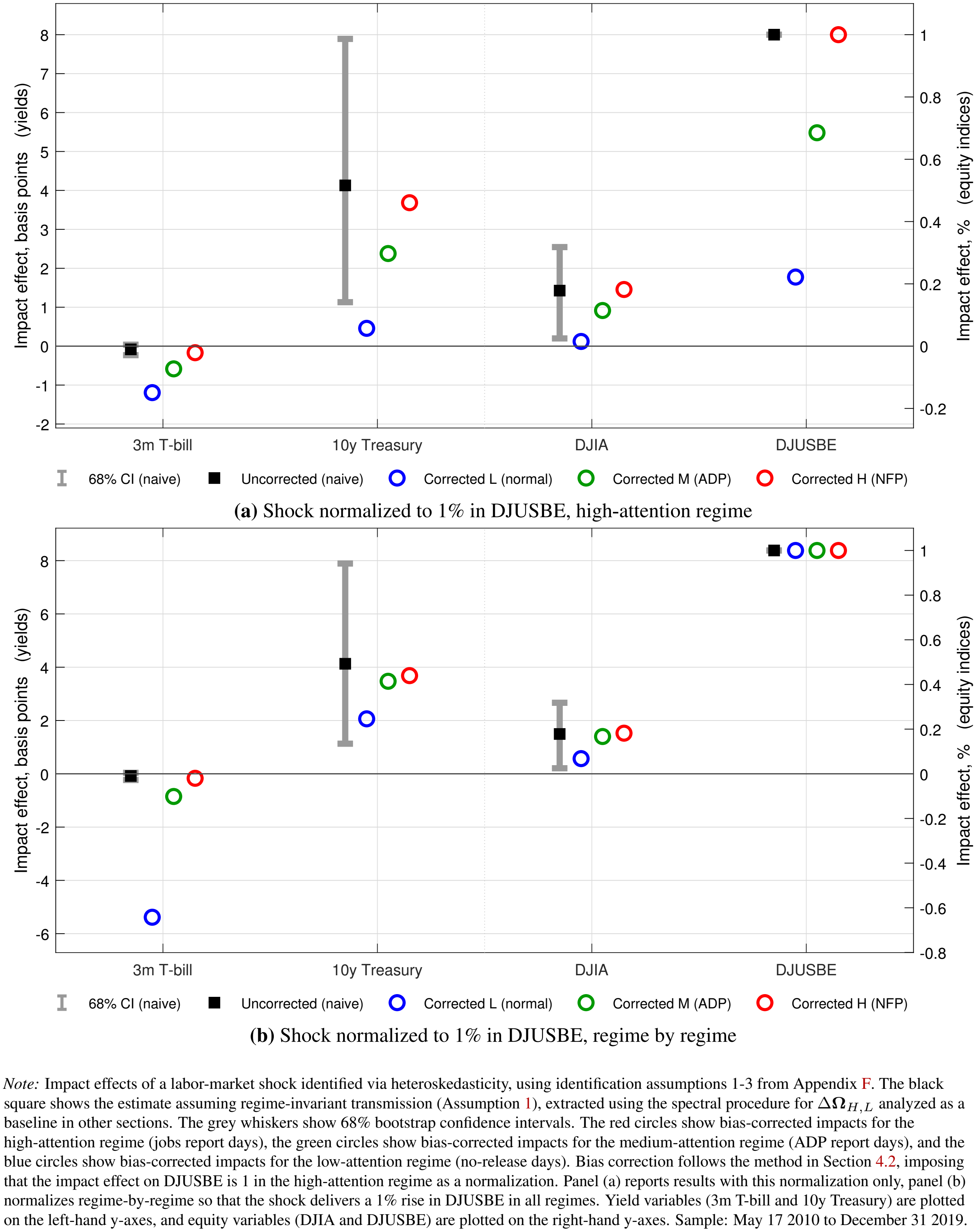
First, stronger real activity raises demand for transaction balances and near-money assets, of which Treasury bills are a prime example (Krishnamurthy and Vissing-Jorgensen, 2012). Second, generic stock market improvements lead to new money coming into asset markets from retail investors. Since this often sits in brokerage accounts for a short time before being invested, that implies a temporary inflow to money-market funds, which mechanically buy T-bills, putting short-run downward pressure on the rate. Third, greater risk appetite supports leveraged positions, raising demand for bills as collateral. Each of these channels operates on the short rate specifically, and does not rely on beliefs about payrolls: such updates to expectations then move both short and long rates up through expectations of monetary policy tightening.

This exercise is intended principally as a demonstration of the application of correction method 2, introduced in Section 4.2. A more detailed investigation of the transmission of labor-market shocks could explore the possible sources of mechanical effects on the T-bill rate, perhaps by extending the list of variables used in the VAR, or extending the sample. I leave this for future work.

Correction method 2 in the application to Wright (2012). In principle, one might think of also applying correction method 2 to the monetary policy shock analysis of Section 5. To do this requires at least three variance regimes, beyond the two-regime split into FOMC days and non-FOMC days common in the literature. Footnote 22 explains why one candidate, separating high-variance days into two regimes based on the realized shocks, is inappropriate. Alternatives, such as creating a middle regime from days of FOMC minutes releases or major pre-scheduled speeches, or dividing high-variance days based on whether they were accompanied by a press conference, respect Assumption 3 but fail the same diagnostic checks used above: trace orderings, variances projected onto candidate direction vectors, and the dominance of a positive eigenvalue.[36] In the ZLB sample studied in Section 5, at least, it therefore seems that separating three distinct variance regimes for monetary policy shocks is a substantial challenge.

[36]In the minutes release days candidate regime, for example, the trace of $\mathbf{\Omega}^{(M)}$ is 0.019, close to (and even slightly below) trace$(\mathbf{\Omega}^{(L)}) = 0.020$. Projecting these onto three candidate directions ($\hat{b}_1$ and the corrected impact vectors from the application of method 1 in Section 5.3) reveals the same pattern: the shock variance in the proposed M regime is hard to distinguish from that in the L regime, but if anything it is slightly lower, and likewise the leading eigenvalue of $\Delta\mathbf{\Omega}_{M,L}$ is negative rather than positive. The required $\sigma_H^2 > \sigma_M^2 > \sigma_L^2$ ordering is therefore unlikely to hold.

Figure 9: Impact effects of a labor-market shock with and without the correction for endogenous attention



(a) Shock normalized to 1% in DJUSBE, high-attention regime

(b) Shock normalized to 1% in DJUSBE, regime by regime

Note: Impact effects of a labor-market shock identified via heteroskedasticity, using identification assumptions 1-3 from Appendix F. The black square shows the estimate assuming regime-invariant transmission (Assumption 1), extracted using the spectral procedure for $\Delta\boldsymbol{\Omega}_{H,L}$ analyzed as a baseline in other sections. The grey whiskers show 68% bootstrap confidence intervals. The red circles show bias-corrected impacts for the high-attention regime (jobs report days), the green circles show bias-corrected impacts for the medium-attention regime (ADP report days), and the blue circles show bias-corrected impacts for the low-attention regime (no-release days). Bias correction follows the method in Section 4.2, imposing that the impact effect on DJUSBE is 1 in the high-attention regime as a normalization. Panel (a) reports results with this normalization only, panel (b) normalizes regime-by-regime so that the shock delivers a 1% rise in DJUSBE in all regimes. Yield variables (3m T-bill and 10y Treasury) are plotted on the left-hand y-axes, and equity variables (DJIA and DJUSBE) are plotted on the right-hand y-axes. Sample: May 17 2010 to December 31 2019. The VAR is estimated with no lags on daily data.